

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2023

Analysis Performed by : SL User

Job File : ...\\401-E-32003-AK.pvdb

Date of Analysis : Jul 4, 2026 11:06am

PV Elite 27, January 2025

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2023

Diameter Spec : 500.000 x 700.000 mm ID
 Vessel Design Length, Tangent to Tangent 2942.85 mm

Distance of Bottom Tangent above Grade 300.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 0.00 mm

Shell Side Design Temperature 80 °C
 Channel Side Design Temperature 150 °C

Shell Side Design Pressure 2300.000 KPa.
 Channel Side Design Pressure 2300.000 KPa.

Shop Shell Side Test Pressure 3736.363 KPa.
 Field Shell Side Test Pressure 4465.586 KPa.
 Shop Channel Side Test Pressure 3064.784 KPa.
 Field Channel Side Test Pressure 3296.412 KPa.

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normalized	Impact Tested
Shell	SA-516 60	...	...	K02100	No	No
Shell	SA-516 70	...	...	K02700	No	No
Head	SA-516 70	...	...	K02700	No	No
Cone	SA-516 60	...	...	K02100	No	No
Flange	SA-350 LF1	1	...	K03009	No	Yes
Nozzle	SA-333 6	...	...	K03006	No	Yes
Nozzle	SA-106 B	...	...	K03006	No	No
Re-Pad	SA-516 70	...	...	K02700	No	No
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-334 6	...	...	K03006	No	Yes
Tubesheet	SA-350 LF2	1	...	K03011	No	Yes
Exp. Jt.	SA-350 LF1	1	...	K03009	No	Yes
Flg Bolting	SA-320 L7	...	<= 65	G41400	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Bottom Ch Flg	2321.605	0.00	2850.000	1.5000	N/A	No
Bottom Ch Shell-1	2321.035	0.00	3887.000	1.5000	N/A	No
Cone	2320.692	0.00	3411.000	1.5000	N/A	No
Bottom Ch Shell-2	2315.992	0.00	2796.000	1.5000	N/A	No
LHS Bottom Ch Flg	2315.649	0.00	2747.000	1.5000	N/A	No
Main Shell	2308.542	103.42	3281.000	1.5000	N/A	No
Ch Top Flg	2305.014	103.42	2757.000	1.5000	N/A	No
Top Ch Shell	2304.294	103.42	3285.000	1.5000	N/A	No
Top Dish End	2301.549	103.42	3150.000	1.5000	No	No

Liquid Level: 2948.80 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 0.700

Element Types and Properties:



Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	89.3	83.0	70.0	63.0	56.7	56.7	1.00	1.00
Cylinder	139.3	50.0	10.0	10.0	6.5	...	1.00	1.00
Conical	824.4	685.0	12.0	12.0	8.6	...	1.00	1.00
Cylinder	874.4	50.0	10.0	10.0	8.5	...	1.00	1.00
Body Flg	985.7	105.0	93.0	85.0	78.1	74.0	1.00	1.00
Cylinder	2305.7	1245.0	10.0	10.0	7.4	4.4	1.00	1.00
Body Flg	2492.0	105.0	108.0	85.0	78.0	74.0	1.00	1.00
Cylinder	2892.0	400.0	10.0	10.0	7.4	3.7	1.00	1.00
Ellipse	2942.8	50.8	11.5	9.5	7.3	3.4	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N-13-Shell Inl	WRC-297	0.986	Shell	Passed
N14-Shell Outl	WRC-297	0.986	Shell	Passed
N9-Tube Outlet	WRC-107/537	0.734	n/a	Passed

Weights:

Fabricated - Bare w/o removable internals	2249.4 kg.
Shop Test - Fabricated + water (full)	3242.4 kg.
Shipping - Fab. + rem. intls.+ shipping app.	2494.9 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2494.9 kg.
Empty - Fab. + intls. + details + wghts.	2494.9 kg.
Operating - Empty + operating liquid (no ca)	3192.2 kg.
Field Test - Empty weight + test medium (full)	3487.9 kg.

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N-13-Shell Inle	3281	...	...	OK	4.92	OK	No Calc[*]
N14-Shell Outle	2945	OK	...	OK	7.42	OK	Passed
K1A-Liquid Leve	3281	...	...	OK	6.30	OK	No Calc[*]
K1B-Liquid Leve	3281	...	...	OK	6.30	OK	No Calc[*]
N9-Tube Outlet	3150	...	...	OK	4.92	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2945 Nozzle : N14-Shell Outle [Shellside]
 Minimum MAWP Nozzles : 3150 Nozzle : N9-Tube Outlet [Tubeside]

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
60	N-13-Shell Inlet	1210.700	180.000	82.352
60	N14-Shell Outlet	2120.700	0.000	298.659
60	K1A-Liquid Level	1820.700	0.000	153.320
60	K1B-Liquid Level	1210.700	0.000	153.320
90	N9-Tube Outlet	0.000	90.000	91.700

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N-13-Shell Inlet	50 mm	XXS	WNF	60.3	11.074	...	...	161	600
N9-Tube Outlet	50 mm	160	LWN	60.3	8.738	...	...	160	600
K1A-Liquid Level	80 mm	80	WNF	88.9	7.620	...	...	162	300
K1B-Liquid Level	80 mm	80	WNF	88.9	7.620	...	...	162	300
N14-Shell Outlet	150 mm	80	WNF	168.3	10.973	228.27	10.00	170	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N-13-Shell	SA-333 6	10.000	10.000	...	...	...
N9-Tube Out	SA-106 B	...	10.000	...	...	...
K1A-Liquid	SA-333 6	7.620	10.000	...	...	...
K1B-Liquid	SA-333 6	7.620	10.000	...	...	...
N14-Shell O	SA-333 6	10.000	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N-13-Shell Inlet	1210.700	180.0	150.00	0.00	Main Shell
N9-Tube Outlet	...	90.0	150.00	0.00	Top Dish End
K1A-Liquid Level	1820.700	0.0	150.00	0.00	Main Shell
K1B-Liquid Level	1210.700	0.0	150.00	0.00	Main Shell
N14-Shell Outlet	2120.700	0.0	150.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	WELD NECK FLANGE: 70.0mm THK X 650.0mm OD	SA-350 LF1
1	CYLINDER: 10.0mm THK X 500.0mm ID X 50.0mm	SA-516 60
1	CYLINDER: 10.0mm THK X 700.0mm ID X 50.0mm	SA-516 60
1	WELD NECK FLANGE: 93.0mm THK X 890.0mm OD	SA-350 LF1
1	CYLINDER: 10.0mm THK X 700.0mm ID X 1245.0mm	SA-516 70
1	WELD NECK FLANGE: 108.0mm THK X 890.0mm OD	SA-350 LF1
1	CYLINDER: 10.0mm THK X 700.0mm ID X 400.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 11.5mm THK X 700.0mm ID X 50.8mm	SA-516 70
1	INSULATION: 83mm X 185mm THK	Cold Insulation
2	INSULATION: 50mm X 185mm THK	Cold Insulation
1	INSULATION: 685mm X 185mm THK	Cold Insulation
2	INSULATION: 105mm X 185mm THK	Cold Insulation
2	TUBESHEET: 890.0mm X 75.0mm THK	SA-350 LF2
316	TUBES: 1400.0mm X 25.0mm DIA X 2.0mm THK	SA-334 6
1	THK EXP JT: 900.0mm DIA X 10.0mm THK	SA-350 LF1
1	INSULATION: 1245mm X 185mm THK	Cold Insulation
1	INSULATION: 400mm X 185mm THK	Cold Insulation
1	INSULATION: 225mm X 185mm THK	Cold Insulation
1	GASKET: 552.0mm OD X 526.0mm ID	...
28	BODY FLANGE BOLTS: 20.0mm DIA	SA-320 L7
28	NUTS FOR BODY FLANGE BOLTS: 20.0mm DIA	...
2	GASKET: 764.0mm OD X 732.0mm ID	...
56	BODY FLANGE BOLTS: 27.0mm DIA	SA-320 L7
56	NUTS FOR BODY FLANGE BOLTS: 27.0mm DIA	...
2	CONICAL SEGMENT 12.0mm THK X 873.5mm X 685.0mm X 623.9mm	SA-516 60
1	CONICAL SEGMENT 12.0mm THK X 452.0mm X 685.0mm X 322.9mm	SA-516 60
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Main Shell [8]		B	-29	-36	-29	0.872	10.000	1.00	Yes
N-13-Shell Inl[1]		B	-8	-104	-29	0.050	9.690	1.00	Yes
Nozzle Flg [5]		A	-18	-104					
N14-Shell Outl[1]		B	-29	-46	-29	0.697	10.000	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
K1A-Liquid Lev[1]		B	-8	-104	-29	0.147	6.668	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
K1B-Liquid Lev[1]		B	-8	-104	-29	0.147	6.668	1.00	Yes
Nozzle Flg [5]		A	-18	-48					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	Yes
Tubesheet: SS[13]		B	-26	-31	-29	0.905	18.750	1.00	Yes
Warmest MDMT:			-8	-31					
LHS Bottom Ch[11]	!		-29	-29	-29	0.603	10.000	1.00	Yes
LHS Bottom Ch[11]	!		-29	-29	-29	0.800	10.000	1.00	Yes
Ch Top Flg [11]	!		-29	-29	-29	0.717	10.000	1.00	Yes
Bottom Ch Shel[8]	C		-44	-48	-29	0.598	10.000	1.00	Yes
Cone [8]	C		-39	-48	-29	0.684	12.000	1.00	Yes
Bottom Ch Shel[8]	C		-44	-48	-29	0.836	10.000	1.00	Yes
Top Ch Shell [8]	B		-29	-45	-29	0.714	10.000	1.00	Yes
Top Dish End [10]	B		-29	-43	-29	0.748	9.500	1.00	Yes
Top Dish End [7]	B		-24	-46	-29	0.607	11.500	1.00	Yes
N9-Tube Outlet[1]	B		-29	-104	-29	0.074	7.645	1.00	Yes
Nozzle Flg [5]	A		-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	Yes
Tubesheet: CS[14]	!		-46	-46		0.905	18.750	1.00	Yes
Warmest MDMT:			-18	-29					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-31.0		-38.0		Fail			
Channel/Tube		-29.0		-38.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.



Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2300	KPa.
Channel Side Design Pressure	2300	KPa.
Shell Side Design Temperature	80.0	°C
Channel Side Design Temperature	150.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-38.0	°C
MDMT (CET), Tube Side	-38.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175

0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	
Damping Factor	2.000	



User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Bottom Ch Flg
Distance "FROM" to "TO"	83 mm
Flange Inside Diameter	500 mm
Element Thickness	63 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	70 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2300 KPa.



Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF1	[Impact Tested]
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	184.5	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	183	N./mm ²
UCS-66 Chart Curve Designation	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K03009	
Class / Thickness / Grade	1	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	83	mm
Liquid Density	0.0006989	kg./cm ³

Element From Node	10	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	83	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³

Element#	2/9	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Bottom Ch Shell-1	
Distance "FROM" to "TO"	50	mm
Inside Diameter	500	mm
Element Thickness	10	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2300	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 60	
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	198	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	195	N./mm ²
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	



Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	50	mm
Liquid Density	0.0006989	kg./cm ³

Element From Node	20	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	50	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³

Element#	3/9	
Element From Node	30	
Element To Node	40	
Element Type	Conical	
Description	Cone	
Distance "FROM" to "TO"	685	mm
Inside Diameter	500	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2300	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	700	mm
Design Length of Cone	685	mm
Half Apex Angle of Cone	8.3056755	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	685	mm
Liquid Density	0.0006989	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	685	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³

Element#	4/9	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Bottom Ch Shell-2	
Distance "FROM" to "TO"	50	mm
Inside Diameter	700	mm
Element Thickness	10	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2300	KPa.
Design Temperature Internal Pressure	150	°C
Design External Pressure	0	KPa.



Design Temperature External Pressure 150 °C
Effective Diameter Multiplier 1.2
Material Name SA-516 60
Efficiency, Longitudinal Seam 1.0
Efficiency, Circumferential Seam 1.0
Weld is pre-Heated No

Element From Node 40
Detail Type Liquid
Detail ID Liquid: 40
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Liquid 50 mm
Liquid Density 0.0006989 kg./cm³

Element From Node 40
Detail Type Insulation
Detail ID Cold Insulation
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Insulation 50 mm
Thickness of Insulation 185 mm
Density 0.00015 kg./cm³

Element# 5/9
Element From Node 50
Element To Node 60
Element Type Flange
Description LHS Bottom Ch Flg
Distance "FROM" to "TO" 105 mm
Flange Inside Diameter 700 mm
Element Thickness 85 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 93 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 2300 KPa.
Design Temperature Internal Pressure 150 °C
Design External Pressure 0 KPa.
Design Temperature External Pressure 150 °C
Effective Diameter Multiplier 1.2
Material Name SA-350 LF1 [Impact Tested]
Allowable Stress, Ambient 118 N./mm²
Allowable Stress, Operating 118 N./mm²
Allowable Stress, Hydrotest 184.5 N./mm²
Material Density 0.00775 kg./cm³
P Number Thickness 32.004 mm
Yield Stress, Operating 183 N./mm²
UCS-66 Chart Curve Designation Impact Tested
External Pressure Chart Name CS-2
UNS Number K03009
Class / Thickness / Grade 1
Product Form Forgings
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 50
Detail Type Liquid
Detail ID Liquid: 50
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Liquid 105 mm
Liquid Density 0.0009984 kg./cm³

Element From Node 50
Detail Type Insulation
Detail ID Cold Insulation
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Insulation 105 mm
Thickness of Insulation 185 mm
Density 0.00015 kg./cm³



Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1245 mm
Inside Diameter	700 mm
Element Thickness	10 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2300 KPa.
Design Temperature Internal Pressure	80 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	80 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	243 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1245 mm
Liquid Density	0.0006989 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Cold Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1245 mm
Thickness of Insulation	185 mm
Density	0.00015 kg./cm ³
Element From Node	60
Detail Type	Nozzle
Detail ID	N-13-Shell Inlet
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	50 mm
Nozzle Schedule	XXS
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.0758 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 [Impact Tested]
Element From Node	60
Detail Type	Nozzle
Detail ID	N14-Shell Outlet
Dist. from "FROM" Node / Offset dist	1060 mm
Nozzle Diameter	150 mm
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0) 29.85 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 [Impact Tested]

Element From Node 60
Detail Type Nozzle
Detail ID K1A-Liquid Level
Dist. from "FROM" Node / Offset dist 760 mm
Nozzle Diameter 80 mm
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 10.428 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 [Impact Tested]

Element From Node 60
Detail Type Nozzle
Detail ID K1B-Liquid Level
Dist. from "FROM" Node / Offset dist 150 mm
Nozzle Diameter 80 mm
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 0.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 10.428 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 [Impact Tested]

Element# 7/9
Element From Node 70
Element To Node 80
Element Type Flange
Description Ch Top Flg
Distance "FROM" to "TO" 105 mm
Flange Inside Diameter 700 mm
Element Thickness 85 mm
Internal Corrosion Allowance 1.5 mm
Nominal Thickness 108 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 2300 KPa.
Design Temperature Internal Pressure 80 °C
Design External Pressure 103.42 KPa.
Design Temperature External Pressure 80 °C
Effective Diameter Multiplier 1.2
Material Name SA-350 LF1 [Impact Tested]
Allowable Stress, Ambient 118 N./mm²
Allowable Stress, Operating 118 N./mm²
Allowable Stress, Hydrotest 184.5 N./mm²
Material Density 0.00775 kg./cm³
P Number Thickness 32.004 mm
Yield Stress, Operating 192.43 N./mm²
UCS-66 Chart Curve Designation Impact Tested
External Pressure Chart Name CS-2
UNS Number K03009
Class / Thickness / Grade 1
Product Form Forgings
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 70
Detail Type Liquid
Detail ID Liquid: 70
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Liquid 105 mm
Liquid Density 0.0006989 kg./cm³



Element From Node	70	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	105	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³

Element#	8/9	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Top Ch Shell	
Distance "FROM" to "TO"	400	mm
Inside Diameter	700	mm
Element Thickness	10	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2300	KPa.
Design Temperature Internal Pressure	80	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	243	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	400	mm
Liquid Density	0.0006989	kg./cm ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	400	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³

Element#	9/9	
Element From Node	90	
Element To Node	100	
Element Type	Elliptical	
Description	Top Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	700	mm
Element Thickness	9.5	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	11.5	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2300	KPa.



Design Temperature Internal Pressure	80	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	80	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	225.8	mm
Liquid Density	0.0006989	kg./cm ³
Element From Node	90	
Detail Type	Insulation	
Detail ID	Cold Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	225.8	mm
Thickness of Insulation	185	mm
Density	0.00015	kg./cm ³
Element From Node	90	
Detail Type	Nozzle	
Detail ID	N9-Tube Outlet	
Dist. from "FROM" Node / Offset dist	25.4	mm
Nozzle Diameter	50	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.6225	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Bottom Ch F	...	89.35	...	83	...
Bottom Ch Shell	...	139.35	...	50	...
Cone	...	824.35	...	685	...
Bottom Ch Shell	...	874.35	...	50	...
LHS Bottom Ch F	...	985.7	...	105	...
Main Shell	...	2305.7	...	1245	...
Ch Top Flg	...	2492.05	...	105	...
Top Ch Shell	...	2892.05	...	400	...
Top Dish End	...	2942.85	...	50.8	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

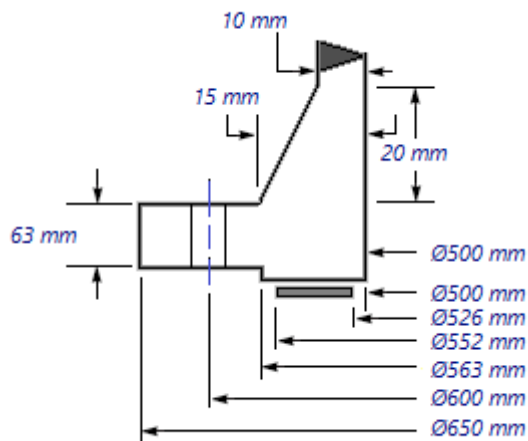
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Flange Input Data Values

Description: FLANGE :

LHS Bottom Ch Flg

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	2321.60	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	500.000	mm
Flange Outside Diameter	A	650.000	mm
Flange Thickness	t	63.0000	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-350 LF1	
Flange Material UNS number		K03009	
Flange Allowable Stress At Temperature	Sfo	118.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	118.00	N./mm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	600.000	mm
Nominal Bolt Diameter	a	20.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	563.000	mm
Flange Face Inside Diameter	Fid	500.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	552.000	mm
Gasket Inside Diameter	Gi	526.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm



ASME Code, Section VIII Division 1, 2023

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2322 \cdot (500/2 + 1.5)) / (118 \cdot 1 - 0.6 \cdot 2322) + Ca$$

= 6.5077 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 8.5) / (252 + 0.6 \cdot 8.5)$$

$$= 3908.510 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 10) / (250 + 0.6 \cdot 10)$$

$$= 4609.024 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	61.500 mm
Corroded Flange ID, Bcor = B+2*ci	503.000 mm
Corroded Large Hub, g1Cor = g1-ci	13.500 mm
Corroded Small Hub, g0Cor = go-ci	8.500 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	35.000 mm
Gasket Contact Width, N = (Go - Gi)/2	13.000 mm
Basic Gasket Width, bo = N/2	6.500 mm
Effective Gasket Width, b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter, G = Go - 2 * b	539.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 539^2 \cdot 2322$$

$$= 54049.965 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 539 \cdot 3 \cdot 2322$$

$$= 15457.527 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 503^2 \cdot 2322$$

$$= 47044.703 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 54050 - 47045$$

$$= 7005.264 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(54050 + 15458 + 0, 0)$$

$$= 69507.492 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 539 + 0 \cdot 0$$

$$= 76508.461 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(69507/172, 76508/172)$$

$$= 43.623 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 20 + 6 \cdot 61.5/(3 + 0.5)$$

$$= 145.429 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 600 \cdot \sin(\pi/28)$$

$$= 67.179 \text{ mm}$$



ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{67.2}{(2 \cdot 20 + 61.5)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	43.623	60.774	
Radial Distance between Hub and Bolts:	28.200	35.000	
Radial Distance between Bolts and Edge:	22.500	25.000	
Circ. Spacing between the Bolts:	46.800	67.179	145.429

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 60.8 \cdot 172 / (68.9 \cdot \pi(552 + 526))$$

$$= 4.476 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa (Am + Ab) / 2$$

$$= 172 (43.6 + 60.8) / 2$$

$$= 91548.66 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wm1 - H$$

$$= 69507 - 54050$$

$$= 15457.53 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (600 - 539) / 2$$

$$= 30.4245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (35 + 13.5 + 30.4) / 2$$

$$= 39.4623 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 35 + (13.5 / 2)$$

$$= 41.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	47045.	41.7500	1.0000	1964.
Face Pressure, Mt	7005.	39.4623	1.0000	276.
Gasket Load, Mg	15458.	30.4245	1.0000	470.
Gasket Seating, Matm	91549.	30.4245	1.0000	2785.

Total Moment for Operation, Mop 2711. Kg-m.
 Total Moment for Gasket seating, Matm 2785. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 65.387 mm
 Hub Ratio, h/h0 = HL / H0 0.306
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.588

Flange Factors for Integral Flange:

Factor F 0.876
 Factor V 0.352
 Factor f 1.277

Factors from Figure 2-7.1
 T = 1.801 K = 1.292
 U = 8.480



Y = 7.717 Z = 3.986
 e = 0.0134 1/mm d = 0.11393E+06 mm³
 Stress Factors Alpha = 1.824
 Beta = 2.098 Gamma = 1.013
 Delta = 2.042 Lambda = 3.054

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g1^2)$
 $= (1.28 \cdot 2711 / 503) / (3.05 \cdot 13.5^2)$
 $= 121.24 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g1^2)$
 $= (1.28 \cdot 2785 / 503) / (3.05 \cdot 13.5^2)$
 $= 124.57 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.1 \cdot 2711 / 503) / (3.05 \cdot 61.5^2)$
 $= 9.60 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.1 \cdot 2785 / 503) / (3.05 \cdot 61.5^2)$
 $= 9.86 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (7.72 \cdot 2711 / (61.5^2 \cdot 503)) - 3.99 \cdot 9.6$
 $= 69.58 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (7.72 \cdot 2785 / (61.5^2 \cdot 503)) - 3.99 \cdot 9.86$
 $= 71.49 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (121 + \max(9.6, 69.6)) / 2$
 $= 95.41 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (125 + \max(9.86, 71.5)) / 2$
 $= 98.03 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 69507 / 60.8$
 $= 112.16 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (76508 / 60.8)$
 $= 123.46 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	121.24	177.00	124.57	177.00
Radial Flange	9.60	118.00	9.86	118.00
Tangential Flange	69.58	118.00	71.49	118.00
Maximum Average	95.41	118.00	98.03	118.00
Bolting	112.16	172.00	123.46	172.00

Minimum Required Flange Thickness 56.667 mm

Estimated M.A.W.P. (Operating) 2871 KPa.

Estimated Finished Weight of Flange at given Thk. 69.3 kg.
Estimated Unfinished Weight of Forging at given Thk 87.2 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 2785 / 1 \cdot 9807 \cdot 0.35 / (2.44 \cdot 202713 \cdot 8.5^2 \cdot 65.4 \cdot 0.3)$
= 0.715 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 2711 / 1 \cdot 9807 \cdot 0.35 / (2.44 \cdot 195074 \cdot 8.5^2 \cdot 65.4 \cdot 0.3)$
= 0.723 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 2785 / 1 \cdot 9807 \cdot 0.35 / (3.05 \cdot 202713 \cdot 8.5^2 \cdot 65.4 \cdot 0.3)$
= 0.570 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 2711 / 1 \cdot 9807 \cdot 0.35 / (3.05 \cdot 195074 \cdot 8.5^2 \cdot 65.4 \cdot 0.3)$
= 0.577 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

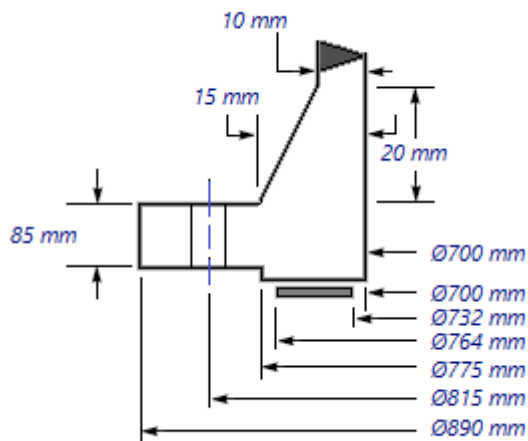
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Flange Input Data Values

Description: FLANGE :

LHS Bottom Ch Flg

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	2315.65	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	890.000	mm
Flange Thickness	t	85.0000	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-350 LF1	
Flange Material UNS number		K03009	
Flange Allowable Stress At Temperature	Sfo	118.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	118.00	N./mm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	815.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	775.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	764.000	mm
Gasket Inside Diameter	Gi	732.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2316 \cdot (700/2 + 1.5)) / (118 \cdot 1 - 0.6 \cdot 2316) + Ca$$

= 8.4806 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 2812.461 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 10) / (350 + 0.6 \cdot 10)$$

$$= 3314.354 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	83.500 mm
Corroded Flange ID, Bcor = B+2*ci	703.000 mm
Corroded Large Hub, g1Cor = g1-ci	13.500 mm
Corroded Small Hub, g0Cor = go-ci	8.500 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	42.500 mm

Gasket Contact Width, N = (Go - Gi)/2	16.000 mm
Basic Gasket Width, bo = N/2	8.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter, G = Go - 2 * b	749.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 750^2 \cdot 2316$$

$$= 104252.516 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 750 \cdot 3 \cdot 2316$$

$$= 23785.715 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 703^2 \cdot 2316$$

$$= 91657.891 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 104253 - 91658$$

$$= 12594.629 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(104253 + 23786 + 0, 0)$$

$$= 128038.234 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 750 + 0 \cdot 0 \cdot 0$$

$$= 118032.398 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(128038/172, 118032/172)$$

$$= 73.004 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 27 + 6 \cdot 83.5/(3 + 0.5)$$

$$= 197.143 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 815 \cdot \sin(\pi/28)$$

$$= 91.251 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{91.3}{(2 \cdot 27 + 83.5)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	73.004	115.879	
Radial Distance between Hub and Bolts:	38.000	42.500	
Radial Distance between Bolts and Edge:	30.500	37.500	
Circ. Spacing between the Bolts:	65.000	91.251	197.143

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 116 \cdot 172 / (68.9 \cdot \pi(764 + 732))$$

$$= 6.150 \text{ mm}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 25.000 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa (Am + Ab) / 2$$

$$= 172 (73 + 116) / 2$$

$$= 165635.80 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wm1 - H$$

$$= 128038 - 104253$$

$$= 23785.72 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (815 - 750) / 2$$

$$= 32.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (42.5 + 13.5 + 32.6) / 2$$

$$= 44.3137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 42.5 + (13.5 / 2)$$

$$= 49.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	91658.	49.2500	1.0000	4514.
Face Pressure, Mt	12595.	44.3137	1.0000	558.
Gasket Load, Mg	23786.	32.6274	1.0000	776.
Gasket Seating, Matm	165636.	32.6274	1.0000	5404.

Total Moment for Operation, Mop 5848. Kg-m.
 Total Moment for Gasket seating, Matm 5404. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 77.301 mm
 Hub Ratio, h/h0 = HL / H0 0.259
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.588

Flange Factors for Integral Flange:

Factor F 0.883
 Factor V 0.372
 Factor f 1.439

Factors from Figure 2-7.1

T =	1.812	K =	1.266
Y =	8.367	U =	9.194
e =	0.0114 1/mm	Z =	4.318
		d =	0.13807E+06 mm ³

Stress Factors

Beta =	2.272	Alpha =	1.954
Delta =	4.217	Gamma =	1.079
		Lambda =	5.295

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.44 \cdot 5848 / 703) / (5.3 \cdot 13.5^2)$
 $= 121.62 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.44 \cdot 5404 / 703) / (5.3 \cdot 13.5^2)$
 $= 112.39 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.27 \cdot 5848 / 703) / (5.3 \cdot 83.5^2)$
 $= 5.02 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.27 \cdot 5404 / 703) / (5.3 \cdot 83.5^2)$
 $= 4.64 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (8.37 \cdot 5848 / (83.5^2 \cdot 703)) - 4.32 \cdot 5.02$
 $= 76.22 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (8.37 \cdot 5404 / (83.5^2 \cdot 703)) - 4.32 \cdot 4.64$
 $= 70.43 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (122 + \max(5.02, 76.2)) / 2$
 $= 98.92 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (112 + \max(4.64, 70.4)) / 2$
 $= 91.41 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 128038 / 116$
 $= 108.36 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (118032 / 116)$
 $= 99.89 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	121.62	177.00	112.39	177.00
Radial Flange	5.02	118.00	4.64	118.00
Tangential Flange	76.22	118.00	70.43	118.00
Maximum Average	98.92	118.00	91.41	118.00
Bolting	108.36	172.00	99.89	172.00



Minimum Required Flange Thickness 78.130 mm
Estimated M.A.W.P. (Operating) 2762 KPa.
Estimated Finished Weight of Flange at given Thk. 160.6 kg.
Estimated Unfinished Weight of Forging at given Thk 193.1 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5404 / 1 \cdot 9807 \cdot 0.37 / (4.29 \cdot 202713 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.704 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5848 / 1 \cdot 9807 \cdot 0.37 / (4.29 \cdot 195074 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.792 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5404 / 1 \cdot 9807 \cdot 0.37 / (5.3 \cdot 202713 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.571 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5848 / 1 \cdot 9807 \cdot 0.37 / (5.3 \cdot 195074 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.642 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

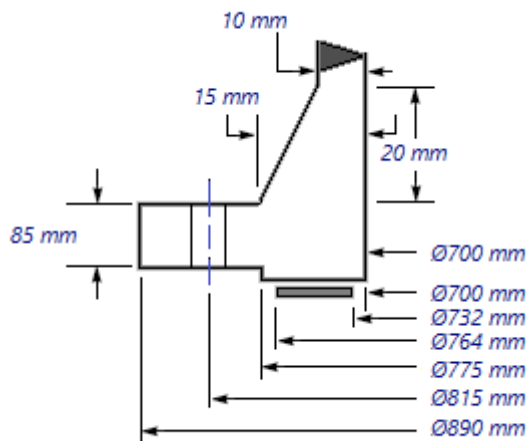
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Flange Input Data Values

Description: New Flange :

Ch Top Flg

Flange Type/Sketch	Integral Weld Neck		
Design Pressure	P	2305.01	KPa.
Design Temperature		80	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	700.000	mm
Flange Outside Diameter	A	890.000	mm
Flange Thickness	t	85.0000	mm
Thickness of Hub at Small End	go	10.0000	mm
Thickness of Hub at Large End	gl	15.0000	mm
Length of Hub	h	20.0000	mm
Flange Material		SA-350 LF1	
Flange Material UNS number		K03009	
Flange Allowable Stress At Temperature	Sfo	118.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	118.00	N./mm ²
Bolt Material		SA-320 L7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	815.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	775.000	mm
Flange Face Inside Diameter	Fid	700.000	mm
Flange Facing Sketch		1, Code Sketch 1a	
Gasket Outside Diameter	Go	764.000	mm
Gasket Inside Diameter	Gi	732.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating		2, Code Column II	
Gasket Thickness	tg	3.1750	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2305 \cdot (700/2 + 1.5)) / (138 \cdot 1 - 0.6 \cdot 2305) + Ca$$

= 7.4310 mm

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 3289.149 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10) / (350 + 0.6 \cdot 10)$$

$$= 3876.109 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	83.500 mm
Corroded Flange ID, Bcor = B+2*ci	703.000 mm
Corroded Large Hub, g1Cor = g1-ci	13.500 mm
Corroded Small Hub, g0Cor = go-ci	8.500 mm
Code R Dimension, R = ((C-Bcor)/2)-g1cor	42.500 mm

Gasket Contact Width, N = (Go - Gi)/2	16.000 mm
Basic Gasket Width, bo = N/2	8.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter, G = Go - 2 * b	749.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 750^2 \cdot 2305$$

$$= 103773.727 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 750 \cdot 3 \cdot 2305$$

$$= 23676.477 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 703^2 \cdot 2305$$

$$= 91236.945 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 103774 - 91237$$

$$= 12536.781 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(103774 + 23676 + 0, 0)$$

$$= 127450.203 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 750 + 0 \cdot 0 \cdot 0$$

$$= 118032.398 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(127450/172, 118032/172)$$

$$= 72.669 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 27 + 6 \cdot 83.5/(3 + 0.5)$$

$$= 197.143 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 815 \cdot \sin(\pi/28)$$

$$= 91.251 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{91.3}{(2 \cdot 27 + 83.5)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	72.669	115.879	
Radial Distance between Hub and Bolts:	38.000	42.500	
Radial Distance between Bolts and Edge:	30.500	37.500	
Circ. Spacing between the Bolts:	65.000	91.251	197.143

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 116 \cdot 172 / (68.9 \cdot \pi(764 + 732))$$

$$= 6.150 \text{ mm}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 25.000 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa (Am + Ab) / 2$$

$$= 172 (72.7 + 116) / 2$$

$$= 165341.78 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wm1 - H$$

$$= 127450 - 103774$$

$$= 23676.47 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (815 - 750) / 2$$

$$= 32.6274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (42.5 + 13.5 + 32.6) / 2$$

$$= 44.3137 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 42.5 + (13.5 / 2)$$

$$= 49.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	91237.	49.2500	1.0000	4494.
Face Pressure, Mt	12537.	44.3137	1.0000	556.
Gasket Load, Mg	23676.	32.6274	1.0000	773.
Gasket Seating, Matm	165342.	32.6274	1.0000	5395.

Total Moment for Operation, Mop 5822. Kg-m.
 Total Moment for Gasket seating, Matm 5395. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 77.301 mm
 Hub Ratio, h/h0 = HL / H0 0.259
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.588

Flange Factors for Integral Flange:

Factor F 0.883
 Factor V 0.372
 Factor f 1.439

Factors from Figure 2-7.1

T =	1.812	K =	1.266
Y =	8.367	U =	9.194
e =	0.0114 1/mm	Z =	4.318
		d =	0.13807E+06 mm ³

Stress Factors

Beta =	2.272	Alpha =	1.954
Delta =	4.217	Gamma =	1.079
		Lambda =	5.295

Longitudinal Hub Stress, Operating [SHo]:
 $= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$
 $= (1.44 \cdot 5822 / 703) / (5.3 \cdot 13.5^2)$
 $= 121.06 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$
 $= (1.44 \cdot 5395 / 703) / (5.3 \cdot 13.5^2)$
 $= 112.19 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.27 \cdot 5822 / 703) / (5.3 \cdot 83.5^2)$
 $= 5.00 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.27 \cdot 5395 / 703) / (5.3 \cdot 83.5^2)$
 $= 4.63 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (8.37 \cdot 5822 / (83.5^2 \cdot 703)) - 4.32 \cdot 5$
 $= 75.87 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (8.37 \cdot 5395 / (83.5^2 \cdot 703)) - 4.32 \cdot 4.63$
 $= 70.31 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (SHo + \max(SRo, STo)) / 2$
 $= (121 + \max(5, 75.9)) / 2$
 $= 98.47 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (SHa + \max(SRa, STa)) / 2$
 $= (112 + \max(4.63, 70.3)) / 2$
 $= 91.25 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= Wm1 / Ab$
 $= 127450 / 116$
 $= 107.86 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (Wm2 / Ab)$
 $= (118032 / 116)$
 $= 99.89 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	121.06	177.00	112.19	177.00
Radial Flange	5.00	118.00	4.63	118.00
Tangential Flange	75.87	118.00	70.31	118.00
Maximum Average	98.47	118.00	91.25	118.00
Bolting	107.86	172.00	99.89	172.00



Minimum Required Flange Thickness 77.953 mm
Estimated M.A.W.P. (Operating) 2762 KPa.
Estimated Finished Weight of Flange at given Thk. 160.6 kg.
Estimated Unfinished Weight of Forging at given Thk 193.1 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5395 / 1 \cdot 9807 \cdot 0.37 / (4.27 \cdot 202713 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.707 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5822 / 1 \cdot 9807 \cdot 0.37 / (4.27 \cdot 199340 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.776 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5395 / 1 \cdot 9807 \cdot 0.37 / (5.3 \cdot 202713 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.570 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 5822 / 1 \cdot 9807 \cdot 0.37 / (5.3 \cdot 199340 \cdot 8.5^2 \cdot 77.3 \cdot 0.3)$
= 0.626 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Bottom Ch Fl		2321.6	70	1.5	500	118	1
Bottom Ch Shell-		2321	10	1.5	500	118	2
Cone		2320.7	12	1.5	700	118	3
Bottom Ch Shell-		2316	10	1.5	700	118	4
LHS Bottom Ch Fl		2315.6	93	1.5	700	118	5
Main Shell		2308.5	10	1.5	700	138	6
Ch Top Flg		2305	108	1.5	700	118	7
Top Ch Shell		2304.3	10	1.5	700	138	8
Top Dish End		2301.5	11.5	1.5	700	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Bottom Ch Fl		2300	2850	3187	63	56.6674	1
Bottom Ch Shell-		2300	3887	4609	10	6.50642	2
Cone		2300	3411	3923	12	8.55598	3
Bottom Ch Shell-		2300	2796	3314	10	8.48166	4
LHS Bottom Ch Fl		2300	2747	2945	85	78.1304	5
Main Shell		2300	3281	3876	10	7.44017	6
Ch Top Flg		2300	2757	2945	85	77.9526	7
Top Ch Shell		2300	3285	3876	10	7.42912	8
Top Dish End		2300	3150	3735	9.5	7.33843	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

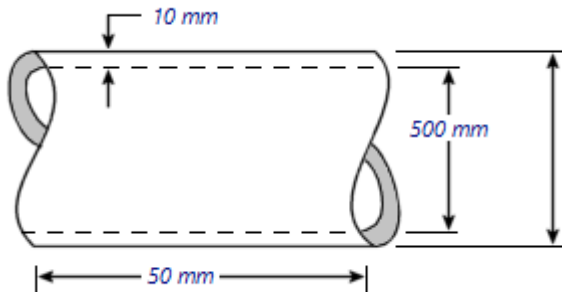
Shell Side MAWP = 2874 KPa.
 Shell Side MAPnc = 3444 KPa.
 Channel Side MAWP = 2358 KPa.
 Channel Side MAPnc = 2541 KPa.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Ch Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	2321.035 KPa.	Int. Design Temperature	150 °C	
Material		SA-516 60	Ext. Design Temperature	150 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02100	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	500.000 mm	Actual Stress	70.1 N./mm ²	
Spec. Min. Thk	t,e, or sa	10.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		816.8 cm ²	SG of Contents	0.700	
Element Volume		9819.2 cm ³	Weight of Contents	6.9 Kg _f	
Empty Weight		6.2 Kg _f	Operating Weight	13.1 Kg _f	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2321 \cdot 252) / (118 \cdot 1 - 0.6 \cdot 2321)$$

$$= 5.0064 + 1.5000 = 6.5064 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 21.035 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 8.5) / (252 + 0.6 \cdot 8.5)$$

$$= 3909 - 21 = 3887 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 10) / (250 + 0.6 \cdot 10)$$

$$= 4609 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2321 \cdot (252 + 0.6 \cdot 8.5)) / (1 \cdot 8.5)$$

$$= 70.072 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 1.961 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 10, tr = 5.09, c = 1.5 mm, E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.6, Temp. Reduction = 23 °C



Min Metal Temp. w/o impact per UCS-66, Curve C -44 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

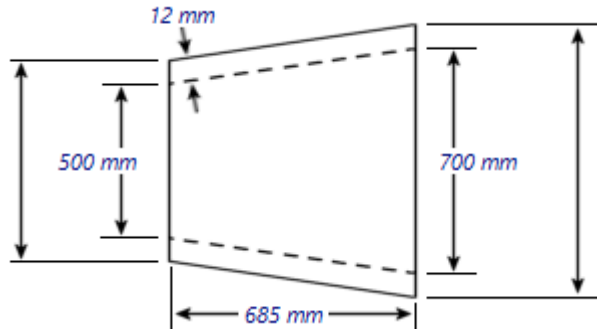
Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	2357.526 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	10.000 mm	Basic MDMT	-44 °C
Stress/Thk. Ratio	0.598	Temperature Difference	4 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Cone

Conical Head/Section Design Information:					
Design Pressure	P	2315.992 KPa.	Int. Design Temperature	150 °C	
Material		SA-516 60	Ext. Design Temperature	150 °C	
Length		685.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02100	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	700.000 mm	Actual Stress	79.8 N./mm ²	
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		13576.3 cm ²	SG of Contents	0.700	
Element Volume		195507.5 cm ³	Weight of Contents	136.8 Kg _f	
Empty Weight		123.8 Kg _f	Operating Weight	260.6 Kg _f	
			To End Diameter	500.000 mm	
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$$
 per Appendix 1-4 (e)

$$= (2316 \cdot 703) / (2 \cdot 0.99 \cdot (118 \cdot 1 - 0.6 \cdot 2316))$$

$$= 7.0560 + 1.5000 = 8.5560 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 Less Operating Hydrostatic Head Pressure of 15.992 KPa.

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 118 \cdot 1 \cdot 10.5 \cdot 0.99) / (703 + 1.2 \cdot 10.5 \cdot 0.99)$$

$$= 3427 - 16 = 3411 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 118 \cdot 1 \cdot 12 \cdot 0.99) / (700 + 1.2 \cdot 12 \cdot 0.99)$$

$$= 3923 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (2316 \cdot (703 + 1.2 \cdot 10.5 \cdot 0.99)) / (2 \cdot 1 \cdot 10.5 \cdot 0.99)$$

$$= 79.750 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.343 %

Note: The Pressure at the Large Diameter is used in the TR calculation.



Minimum Design Metal Temperature Results:

Govrn. thk, tg = 12, tr = 7.18, c = 1.5 mm, $E^* = 1$
Thickness Ratio = $tr * E^* / (tg - c) = 0.68$, Temp. Reduction = 18 °C

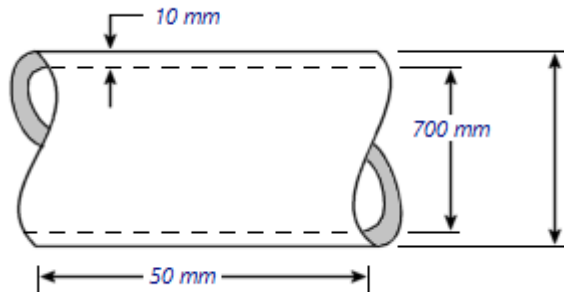
Min Metal Temp. w/o impact per UCS-66, Curve C -39 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	2357.526 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	12.000 mm	Basic MDMT	-39 °C
Stress/Thk. Ratio	0.684	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom Ch Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2315.992 KPa.	Int. Design Temperature	150 °C	
Material		SA-516 60	Ext. Design Temperature	150 °C	
Length		50.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02100	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	700.000 mm	Actual Stress	97.2 N./mm ²	
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		1131.0 cm ²	SG of Contents	0.700	
Element Volume		19245.7 cm ³	Weight of Contents	13.5 Kg	
Empty Weight		8.6 Kg	Operating Weight	22.1 Kg	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2316 \cdot 352) / (118 \cdot 1 - 0.6 \cdot 2316)$$

$$= 6.9817 + 1.5000 = 8.4817 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 15.992 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 2812 - 16 = 2796 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 10) / (350 + 0.6 \cdot 10)$$

$$= 3314 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2316 \cdot (352 + 0.6 \cdot 8.5)) / (1 \cdot 8.5)$$

$$= 97.168 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.408 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 10, t_r = 7.11, c = 1.5 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.84, Temp. Reduction = 9 °C



Min Metal Temp. w/o impact per UCS-66, Curve C -44 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note:

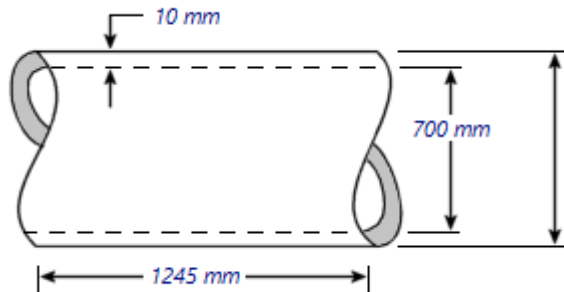
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	2357.526 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	10.000 mm	Basic MDMT	-44 °C
Stress/Thk. Ratio	0.836	Temperature Difference	4 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	2308.542 KPa.	Int. Design Temperature		80 °C
Material		SA-516 70	Ext. Design Temperature		80 °C
Length		1245.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	700.000 mm	Actual Stress		96.9 N./mm ²
Spec. Min. Thk	t _e , or s _a	10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		28161.2 cm ²	SG of Contents		0.700
Element Volume		286063.8 cm ³	Weight of Contents		200.1 Kgf
Empty Weight		215.2 Kgf	Operating Weight		415.4 Kgf

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2309 \cdot 352) / (138 \cdot 1 - 0.6 \cdot 2309)$$

$$= 5.9402 + 1.5000 = 7.4402 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.542 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5)$$

$$= 3289 - 8.54 = 3281 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10) / (350 + 0.6 \cdot 10)$$

$$= 3876 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2309 \cdot (352 + 0.6 \cdot 8.5)) / (1 \cdot 8.5)$$

$$= 96.856 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.408 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 10, t_r = 7.41, c = 1.5 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.87, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -36 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2874.125 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.872	Temperature Difference	7 °C
UCS-66(c) red.	No	Computed Min. Temp.	-36 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.500	720.00	1245.00	84.71	1.7292	0.0009679	84.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*84.3)/(3*84.7) = 1327 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.904	720.00	1245.00	247.92	1.7292	0.0001924	19.23

$$MAEP = (4*B)/(3*(Do/t)) = (4*19.2)/(3*248) = 103 \text{ KPa.}$$

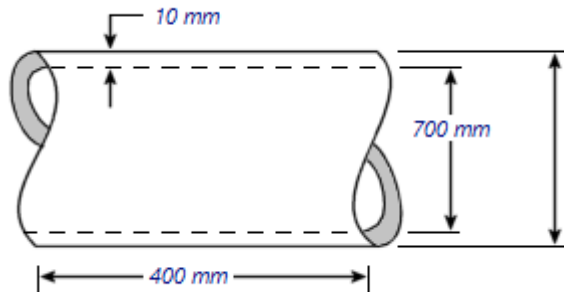
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.500	720.00	32149.87	84.71	44.6526	0.0001554	15.54

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.5)/(3*84.7) = 245 \text{ KPa.}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	2304.294 KPa.	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		400.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	700.000 mm	Actual Stress	96.7 N./mm ²	
Spec. Min. Thk	t _e , or s _a	10.0000 mm	Spec. Nominal Thk.	10.000 mm	
Surface Area		9047.8 cm ²	SG of Contents	0.700	
Element Volume		153965.6 cm ³	Weight of Contents	107.7 Kg	
Empty Weight		69.2 Kg	Operating Weight	176.9 Kg	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2304 \cdot 352) / (138 \cdot 1 - 0.6 \cdot 2304) \\
 &= 5.9291 + 1.5000 = 7.4291 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 4.294 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 8.5) / (352 + 0.6 \cdot 8.5) \\
 &= 3289 - 4.29 = 3285 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 10) / (350 + 0.6 \cdot 10) \\
 &= 3876 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2304 \cdot (352 + 0.6 \cdot 8.5)) / (1 \cdot 8.5) \\
 &= 96.678 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.408 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 10, t_r = 6.07, c = 1.5 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.71, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -45 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2357.526 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.714	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-45 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Top Ch Shell

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.500	720.00	614.13	84.71	0.8530	0.0020621	104.44

$$MAEP = (4*B)/(3*(Do/t)) = (4*104)/(3*84.7) = 1644 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.154	720.00	614.13	334.19	0.8530	0.0002593	25.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.9)/(3*334) = 103 \text{ KPa.}$$

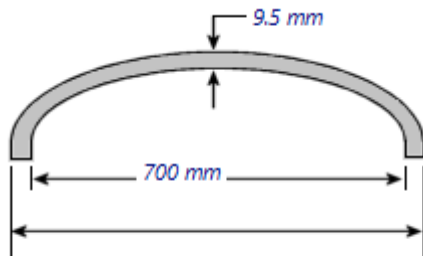
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.500	720.00	16703.21	84.71	23.1989	0.0001565	15.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.6)/(3*84.7) = 246 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2023



Internal Pressure Results for: Top Dish End

Elliptical Head Design Information:					
Design Pressure	P	2301.200 KPa.	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	80 °C	
Length		50.800 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	700.000 mm	Actual Stress	100.8 N./mm ²	
Spec. Min. Thk t _e , or sa		9.5000 mm	Spec. Nominal Thk.	11.500 mm	
Surface Area		6820.2 cm ²	SG of Contents	0.700	
Element Volume		64460.3 cm ³	Weight of Contents	45.1 Kg _f	
Empty Weight		64.1 Kg _f	Operating Weight	109.2 Kg _f	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency	E 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2301 \cdot 703 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2301)$$

$$= 5.8384 + 1.5000 = 7.3384 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 1.201 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8) / (0.99 \cdot 703 + 0.2 \cdot 8)$$

$$= 3151 - 1.2 = 3150 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 9.5) / (1 \cdot 700 + 0.2 \cdot 9.5)$$

$$= 3735 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2301 \cdot (0.99 \cdot 703 + 0.2 \cdot 8)) / (2 \cdot 1 \cdot 8)$$

$$= 100.773 \text{ N./mm}^2$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2301 \cdot 352) / (138 \cdot 1 - 0.6 \cdot 2301) + 1.5$$

$$= 7.421 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 1.201 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 10) / (352 + 0.6 \cdot 10)$$

$$= 3860 - 1.2 = 3859 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (703 / (2 \cdot 176)))^2 / 6$$

$$= 0.994346$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.914 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 9.5$, $t_r = 5.98$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 11.5$, $t_r = 6.07$, $c = 1.5$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.61$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

[Note:](#)
[Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.](#)

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2357.526 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	9.500 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.748	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 8.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.375 mm

EXTERNAL PRESSURE RESULTS for: Top Dish End

Elliptical Head From 90 to 100 Ext. Chart: CS-2 at 80 °C

Elastic Modulus from Chart: CS-2 at 80 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.000	719.00	89.88	0.0015454	97.56



MAEP = $B / (K0 * Do / t) = 97.6 / (0.9 * 89.9) = 1206 \text{ KPa.}$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.862	719.00	386.22	0.0003596	35.95

MAEP = $B / (K0 * Do / t) = 36 / (0.9 * 386) = 103 \text{ KPa.}$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3736.363 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2990.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4470.928 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	3161.538 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3289.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3289.000 KPa.
App 27-4	= M.A.W.P.	2874.125 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3064.784 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2990.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3296.714 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2593.279 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3289.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3289.000 KPa.
App 27-4	= M.A.W.P.	2357.526 KPa.

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom Ch Shell-1	78.6	198.0	0.397	3069.69
Cone	92.4	198.0	0.467	3071.65
Bottom Ch Shell-2	109.4	198.0	0.552	3071.65
Main Shell	133.3	234.0	0.570	3743.22
Top Ch Shell	109.4	234.0	0.467	3071.65
Top Dish End	113.5	234.0	0.485	3071.65

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N-13-Shell Inlet	Nozzle	118.00	118.00	1.000
N14-Shell Outlet	Nozzle	118.00	118.00	1.000
N14-Shell Outlet	Pad	138.00	138.00	1.000
K1A-Liquid Level	Nozzle	118.00	118.00	1.000
K1B-Liquid Level	Nozzle	118.00	118.00	1.000
N9-Tube Outlet	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Bottom Ch Flg	118.00	118.00	1.000
Bottom Ch Shell-1	118.00	118.00	1.000
Cone	118.00	118.00	1.000
Bottom Ch Shell-2	118.00	118.00	1.000
LHS Bottom Ch Flg	118.00	118.00	1.000
Main Shell	138.00	138.00	1.000
Ch Top Flg	118.00	118.00	1.000
Top Ch Shell	138.00	138.00	1.000
Top Dish End	138.00	138.00	1.000

Minimum 1.000



Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000
Exp. Joint Material	118.00	118.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N-13-Shell Inlet	8.70	153.40	0.057
N14-Shell Outlet	27.21	153.40	0.177
K1A-Liquid Level	20.34	153.40	0.133
K1B-Liquid Level	20.34	153.40	0.133
N9-Tube Outlet	9.38	153.40	0.061

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3736.363 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2990.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4465.586 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	3161.538 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3289.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3289.000 KPa.
App 27-4	= M.A.W.P.	2874.125 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3064.784 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2990.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3296.412 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2593.279 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3289.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3289.000 KPa.
App 27-4	= M.A.W.P.	2357.526 KPa.

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Bottom Ch Shell-1	100.3	198.0	0.507	3321.91
Cone	114.1	198.0	0.576	3314.70
Bottom Ch Shell-2	139.1	198.0	0.702	3314.70
Main Shell	187.9	234.0	0.803	4477.79
Top Ch Shell	138.6	234.0	0.592	3302.55
Top Dish End	144.5	234.0	0.617	3298.63

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N-13-Shell Inlet	Nozzle	118.00	118.00	1.000
N14-Shell Outlet	Nozzle	118.00	118.00	1.000
N14-Shell Outlet	Pad	138.00	138.00	1.000
K1A-Liquid Level	Nozzle	118.00	118.00	1.000
K1B-Liquid Level	Nozzle	118.00	118.00	1.000
N9-Tube Outlet	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Bottom Ch Flg	118.00	118.00	1.000
Bottom Ch Shell-1	118.00	118.00	1.000
Cone	118.00	118.00	1.000
Bottom Ch Shell-2	118.00	118.00	1.000
LHS Bottom Ch Flg	118.00	118.00	1.000
Main Shell	138.00	138.00	1.000
Ch Top Flg	118.00	118.00	1.000
Top Ch Shell	138.00	138.00	1.000
Top Dish End	138.00	138.00	1.000

Minimum 1.000



Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000
Exp. Joint Material	118.00	118.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N-13-Shell Inlet	10.41	153.40	0.068
N14-Shell Outlet	32.55	153.40	0.212
K1A-Liquid Level	24.33	153.40	0.159
K1B-Liquid Level	24.33	153.40	0.159
N9-Tube Outlet	10.07	153.40	0.066

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	61.5	No Calc	No Calc
20	30	133	520	8.5	0.013811	122.677
30	40	590.387	724.254	10.5	0.0029491	111.336
40	50	155	720	8.5	0.010049	122.677
50	60	No Calc	...	83.5	No Calc	No Calc
60	70	1245	720	8.5	0.00096791	84.2908
70	80	No Calc	...	83.5	No Calc	No Calc
80	90	614.133	720	8.5	0.0020621	104.438
90	100	No Calc	719	8	0.0015454	97.5563

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	63	56.6674	...	No Calc
20	30	10	No Calc	...	2674
30	40	12	No Calc	...	2129
40	50	10	No Calc	...	1931
50	60	85	74.0156	...	No Calc
60	70	10	4.40418	103.419	1327
70	80	85	74.0156	103.419	No Calc
80	90	10	3.65447	103.419	1644
90	100	9.5	3.36161	103.419	1206

Minimum

1206

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	133	No Calc	No Calc	No Calc
30	40	590.387	590.387	No Calc	No Calc
40	50	155	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	1245	32149.9	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	614.133	16703.2	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	76.6231	17840.1	74.9746	17887.4	7.66231
20	30	6.20897	9819.24	5.29314	9937.42	0.6209
30	40	123.815	195508	108.606	197470	12.3815
40	50	8.64386	19245.7	7.36279	19411	0.86439
50	60	175.36	43802.8	172.465	43868.9	17.536
60	70	215.232	286064	183.334	290180	21.5232
70	80	202.944	49626.1	199.665	49692.2	20.2944
80	90	69.1509	153966	58.9023	155288	6.91509
90	100	64.1239	64460.3	55.7599	65208.1	6.41239
Total		942	840331.06	866	848943.56	94



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	11.403	...	41.5	...	Liquid: 10
10	Insl	6.90821	...	41.5	...	Cold Insulation
20	Liqd	6.86927	...	25	...	Liquid: 20
20	Insl	3.07667	...	25	...	Cold Insulation
30	Liqd	136.772	...	342.5	...	Liquid: 30
30	Insl	47.219	...	342.5	...	Cold Insulation
40	Liqd	13.4638	...	25	...	Liquid: 40
40	Insl	3.94948	...	25	...	Cold Insulation
50	Liqd	40.3913	...	52.5	...	Liquid: 50
50	Insl	10.9327	...	52.5	...	Cold Insulation
60	Liqd	200.123	...	622.5	...	Liquid: 60
60	Insl	98.342	...	622.5	...	Cold Insulation
60	Nozl	7.78339	-425.006	150	...	N-13-Shell Inlet
60	Nozl	32.8345	425.006	1060	...	N14-Shell Outlet
60	Nozl	11.4706	425.006	760	...	K1A-Liquid Level
60	Nozl	11.4706	425.006	150	...	K1B-Liquid Level
70	Liqd	28.2739	...	52.5	...	Liquid: 70
70	Insl	12.6524	...	52.5	...	Cold Insulation
80	Liqd	107.71	...	200	...	Liquid: 80
80	Insl	31.5958	...	200	...	Cold Insulation
90	Liqd	45.0946	...	138.3	...	Liquid: 90
90	Insl	30.8286	...	112.9	...	Cold Insulation
90	Nozl	7.28478	...	300.345	-25.4	N9-Tube Outlet
50	FTsh	298.606	...	145.675	...	Tubes Tubesheet
50	Tube	545.061	...	808.175	...	
50	RTsh	298.606	...	1470.68	...	

Total Weight of Each Detail Type:

Liquid	590.1
Insulation	245.5
Nozzles	70.8
Exchanger Components	1142.3
Liquid in Tubes	107.2
Sum of the Detail Weights	2155.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1036.3	1036.3	1036.3	1036.3	1036.3	1036.3
Nozzles	70.8	70.8	70.8	70.8	70.8	70.8
Exchanger	1142.3	1142.3	1142.3	1142.3	1142.3	1142.3
Insulation	...	...	245.5	245.5	245.5	245.5
Ope. Liquid	...	...	...	...	...	590.1
Tube Ope Lqd	...	...	...	...	...	107.2
Test Liquid	...	839.8	...	...	...	...
Tube Tst Lqd	...	153.2	...	...	...	...
Totals	2249.4	3242.4	2494.9	2494.9	2494.9	3192.2

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2249.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	3242.4 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2494.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2494.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2494.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	3192.2 kg.
Field Test Wt.	- Empty weight + test medium (full)	3487.9 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	780.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	860.2 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	577.6 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	153257.4 cm ³
Weight of Ope Liq in tubes :	107.2 kg.
Weight of Water in tubes :	153.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	3117.25	1413.72
20	30	816.814	785.398
30	40	13576.3	13048.8
40	50	1130.97	1099.56
50	60	5431.66	2485
60	70	28161.2	27379
70	80	5851.06	2814.87
80	90	9047.79	8796.46
90	100	6820.16	6428.68

Total	73953.2	64251.5 cm ²
Total	79.6	69.2 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	91.1936	102.597	109.023	...	91.1936
20	30	9.90654	16.7758	19.7198	...	9.90654
30	40	183.415	320.187	378.803	...	183.415



40	50	13.4577	26.9215	32.6917	...	13.4577
50	60	1346.1	1386.49	1389.88	...	1346.1
60	70	398.657	598.779	684.546	20.3974	398.657
70	80	235.891	264.165	285.487	...	235.891
80	90	107.662	215.372	261.533	...	107.662
90	100	108.65	153.744	173.071	...	108.65

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	2494.93	3085.03	3334.75
20	30	2403.74	2982.44	3225.73
30	40	2393.83	2965.66	3206.01
40	50	2210.42	2645.48	2827.21
50	60	2196.96	2618.55	2794.51
60	70	850.859	1232.06	1404.64
70	80	452.202	633.281	720.091
80	90	216.311	369.116	434.604
90	100	108.65	153.744	173.071

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	20.3974	20.3974	20.3974
20	30	20.3974	20.3974	20.3974
30	40	20.3974	20.3974	20.3974
40	50	20.3974	20.3974	20.3974
50	60	20.3974	20.3974	20.3974
60	70	20.3974	20.3974	20.3974
70	80	...	...	...
80	90	...	...	...
90	100	...	...	...



Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N-13-Shell Inl	9600.0	10210.0	80	600	GR 1.1	6627.6	7772.4	9600.0	7772.4
N14-Shell Outl	4800.0	5110.0	80	300	GR 1.1	3171.0	4029.0	4800.0	4029.0
K1A-Liquid Lev	4800.0	5110.0	80	300	GR 1.1	...	...	...	...
K1B-Liquid Lev	4800.0	5110.0	80	300	GR 1.1	...	...	...	...
N9-Tube Outlet	9600.0	10210.0	80	600	GR 1.1	6627.6	7772.4	9600.0	7772.4

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4029.006 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5110.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 7772.386 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 10210.003 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N-13-Shell Inlet	77.851	0.50	59	163	2866.6	7772.4	Pass
N14-Shell Outlet	196.088	0.50	459	408	2872.9	4029.0	Pass
N9-Tube Outlet	77.851	0.50	59	163	2357.5	7772.4	Pass

Evaluating Flange on Nozzle [N-13-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(59) + 4 \cdot \text{abs}(163)77.851 \leq \pi \cdot 77.851^3 \cdot ((9600.002 - 2866.612) + .500 \cdot 9600.002)$$

$$= 0.10019 + 04 \leq 0.17434 + 04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N-13-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9600.002, 0.500 \cdot 9600.002 + 9600.002 - 1001.9 / (\pi \cdot 77.851^3))$$

$$= \min(9600.002, 7772.386)$$

$$= 7772.386 \text{ KPa.}$$

Evaluating Flange on Nozzle [N14-Shell Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(459) + 4 \cdot \text{abs}(408)196.088 \leq \pi \cdot 196.088^3 \cdot ((4800.004 - 2872.856) + .500 \cdot 4800.004)$$

$$= 0.76596 + 04 \leq 0.10452 + 05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N14-Shell Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4800.004, 0.500 \cdot 4800.004 + 4800.004 - 7659.6 / (\pi \cdot 196.088^3))$$

$$= \min(4800.004, 4029.006)$$

$$= 4029.006 \text{ KPa.}$$

Evaluating Flange on Nozzle [N9-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(59) + 4 \cdot \text{abs}(163)77.851 \leq \pi \cdot 77.851^3 \cdot ((9600.002 - 2357.523) + .500 \cdot 9600.002)$$

$$= 0.10019 + 04 \leq 0.18204 + 04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N9-Tube Outlet [Pmax]:



$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$
$$= \min(9600.002, 0.500 \cdot 9600.002 + 9600.002 - 1001.9 / (\pi \cdot 77.851^3))$$
$$= \min(9600.002, 7772.386)$$
$$= 7772.386 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2020 Metric Edition

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone

Design Internal Pressure, Small End	2320.692	KPa.
Design Internal Pressure, Large End	2315.992	KPa.
Temperature for Internal Pressure	150.0	°C
Design External Pressure	0.000	KPa.
Temperature for External Pressure	150.0	°C

CONE Properties:

Cone Material		SA-516 60	
Cone Allowable Stress at Temperature	Ss	118.00	N./mm ²
Cone Allowable Stress At Ambient		118.00	N./mm ²
Longitudinal Joint Efficiency of Cone		1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	12.0000	mm
Corrosion Allowance for Cone	Cac	1.5000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	500.000	mm
Diameter of Large End of Cone	Dl	700.000	mm
Half Apex Angle for Cone		8.31	degrees
Axial Length of Cone	Lc	685.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 60	
Operating Allowable Stress	Ss	118.00	N./mm ²
Ambient Allowable Stress		118.00	N./mm ²
Joint Efficiency	Es	1.0000	
Actual Thickness	Ts	10.0000	mm
Corrosion Allowance	Cas	1.5000	mm
Axial Length of Attached Element	Ls	133.000	mm

LARGE-END Properties:

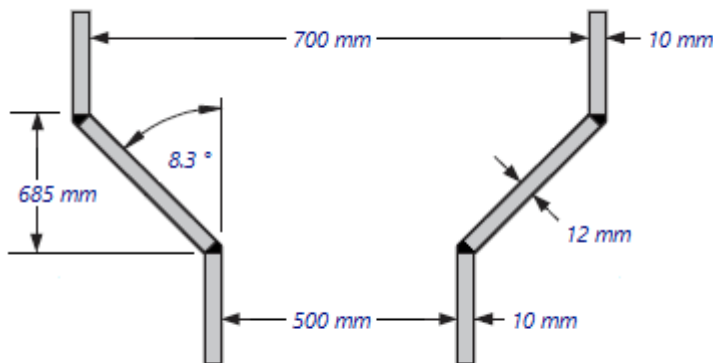
Attached Large End Material Specification		SA-516 60	
Operating Allowable Stress	Sl	118.00	N./mm ²
Ambient Allowable Stress		118.00	N./mm ²
Joint Efficiency	El	1.0000	
Actual Thickness	Tl	10.0000	mm
Corrosion Allowance	Cal	1.5000	mm
Axial Length of Attached Element	Ll	155.000	mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange Yes

Force Arl & Ars calculation, cone -> cylinder Yes



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 150 °C

Elastic Modulus of Cone Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Small Cylinder Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Large Cylinder Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Large End Reinforcement	0.195E+09 KPa. at 150 °C
Elastic Modulus of Small End Reinforcement	0.195E+09 KPa. at 150 °C

Elastic Modulus Data from ASME Section II Part D at 150 °C

Elastic Modulus of Cone Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Small Cylinder Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Large Cylinder Material	0.195E+09 KPa. at 150 °C
Elastic Modulus of Large End Reinforcement	0.195E+09 KPa. at 150 °C
Elastic Modulus of Small End Reinforcement	0.195E+09 KPa. at 150 °C

Forces and Moments on the ends of the Cone:

Since there are no supports, the forces and moments could not be resolved to the large and small ends of the cone.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	13.8293 mm
Maximum Centroid Reinforcement Distance Small End	11.7527 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01963

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 326.6 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-5 equation 1
 $= 326.6 \cdot \sqrt{(2316 / (118 \cdot 1))}$
 $= 45.757$ degrees

Large end actual half apex angle [alpha]: 8.306 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the large end under internal pressure.



Decay Length, Cone Large End:

$$\begin{aligned} &= 2 \cdot \sqrt{(RL (ts - ca))} \\ &= 2 \cdot \sqrt{(352 (10 - 1.5))} \\ &= 109.321 \text{ mm} \end{aligned}$$

Required Area of Reinforcement, Large End, Internal [Arl]:

$$\begin{aligned} &= k \cdot QL \cdot RI / (Sc \cdot E2) \cdot \tan(\alpha) \\ &= 1 \cdot 41.2 \cdot 352 / (118 \cdot 1) \cdot 0.15 \\ &= 1.758 \text{ cm}^2 \end{aligned}$$

Force per Length, Cone Large End, tensile side [QL+]:

$$\begin{aligned} &= P (RI/2) + f1 \\ &= 2300 (352/2) + 0 \\ &= 41.221 \text{ Kgf/mm} \end{aligned}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$\begin{aligned} &= Faxial / (\pi(DI + TI)) + Moment / (\pi(RI + TI/2)^2) \\ &= -0 / (\pi(703 + 10)) + 0 / (\pi(352 + 10/2)^2) \\ &= 0.0 \text{ Kgf/mm} \end{aligned}$$

Force per Length, Cone Large End, compressive side [QL-]:

$$\begin{aligned} &= P (RI/2) + f1 \\ &= 2300 (352/2) + 0 \\ &= 41.221 \text{ Kgf/mm} \end{aligned}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$\begin{aligned} &= Faxial / (\pi(DI + TI)) - Moment / (\pi(RI + TI/2)^2) \\ &= -0 / (\pi(703 + 10)) - 0 / (\pi(352 + 10/2)^2) \\ &= 0.0 \text{ Kgf/mm} \end{aligned}$$

Area of Reinforcement Available in Large end Shell [Ael]:

$$\begin{aligned} &= (tc - tr) \cdot \sqrt{RL \cdot tc / \cos(\alpha)} \\ &= (10.5 - 7.07) \cdot \sqrt{352 \cdot 10.5 / 0.99} \\ &= 2.0947 \text{ cm}^2 \end{aligned}$$

Summary of Reinforcement area, Large end, Internal Pressure:

Area of reinforcement required per App. 1-5(1)	1.7580	cm ²
Area of reinforcement in shell per App. 1-5(2)	2.0947	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01967

Small End Maximum half apex angle w/o reinforcement [δ]:

$$\begin{aligned} &= 89 \cdot \sqrt{(P / (Ss \cdot E1))} \text{ per 1-5 equation 6} \\ &= 89 \cdot \sqrt{(2321 / (118 \cdot 1))} \\ &= 12.482 \text{ degrees} \end{aligned}$$

Small end actual half apex angle [alpha]: 8.306 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the small end under internal pressure.

Decay Length, Cone Small End:

$$\begin{aligned} &= 1.4 \cdot \sqrt{(Rs (ts - ca))} \\ &= 1.4 \cdot \sqrt{(252 (10 - 1.5))} \\ &= 64.730 \text{ mm} \end{aligned}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$\begin{aligned} &= k \cdot QS \cdot Rs / (Sc \cdot E2) \cdot \tan(\alpha) \\ &= 1 \cdot 29.5 \cdot 252 / (118 \cdot 1) \cdot 0.15 \\ &= 0.9 \text{ cm}^2 \end{aligned}$$

Force per Length, Cone Small End, tensile side [QS+]:

$$\begin{aligned} &= P (Rs/2) + f2 \\ &= 2300 (252/2) + 0 \\ &= 29.494 \text{ Kgf/mm} \end{aligned}$$

Axial Load per Unit Circumference excluding Pressure [f2]:
 $= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs + Ts/2)^2)$
 $= -0/(\pi(503+10))+0/(\pi(252+10/2)^2)$
 $= 0.0 \text{ Kg/mm}$

Force per Length, Cone Small End, compressive side [QS-]:
 $= P(Rs/2) + f2$
 $= 2300 (252/2) + 0$
 $= 29.494 \text{ Kg/mm}$

Axial Load per Unit Circumference excluding Pressure [f2]:
 $= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs + Ts/2)^2)$
 $= -0/(\pi(503+10))-0/(\pi(252+10/2)^2)$
 $= 0.0 \text{ Kg/mm}$

Area of Reinforcement Available in Small End Shell [Aes]:
 $= 0.78 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))} \cdot (tc - tr)$ per 1-5 equation 10
 $= 0.78 \cdot \sqrt{(252 \cdot 10.5/0.99)} \cdot (10.5 - 5.06)$
 $= 2.193 \text{ cm}^2$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	0.9000	cm ²
Area of reinforcement in shell per App. 1-5(8)	2.1925	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	68.43	390.00	Small Cyl. Long.
Compres. Stress	0.00	-390.00	Small Cyl. Long. *
Membrane Stress	85.26	-177.00	Small End Tang.
Tensile Stress	50.53	390.00	Cone Longitudinal
Compres. Stress	0.00	-390.00	Cone Longitudinal *
Compres Stress	72.56	-177.00	Cone Tangential
Tensile Stress	103.55	390.00	Large Cyl. Long.
Compres. Stress	-6.42	-390.00	Large Cyl. Long.
Membrane Stress	71.82	-177.00	Large End Tang.
Tensile Stress	75.76	390.00	Cone Longitudinal
Compres. Stress	0.00	-390.00	Cone Longitudinal *
Compres Stress	54.16	-177.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure KPa.	Reason for Failure at this Pressure
MAWP	3411.214	Thickness due to internal pressure, Cone Large End
MAPnc	3923.614	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.



Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	1607.737 mm
Center of Gravity of the Insulation	1674.261 mm
Center of Gravity of the Nozzles	1935.007 mm
Center of Gravity of the Tubesheet(s)	1682.525 mm
Center of Gravity of the Tubes	1682.525 mm
Center of Gravity of Bare Shell New and Cold	1562.293 mm
Center of Gravity of Bare Shell Corroded	1553.464 mm
Vessel CG in the Operating Condition	1633.869 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1638.941 mm

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Input, Nozzle Desc: N-13-Shell Inlet

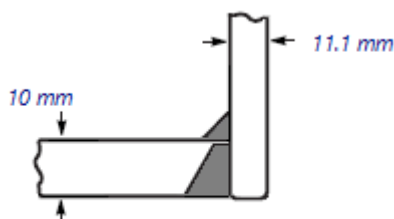
From: 60

Pressure for Reinforcement Calculations	P	2307.513	KPa.
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	1245.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1210.70	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material [Impact Tested]		SA-333 6	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N-13-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 38.176 mm.
 Actual Thickness Used in Calculation 11.074 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2308 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 2308)$
 $= 5.9375$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2308 \cdot 20.6 / (118 \cdot 1 - 0.6 \cdot 2308)$
 $= 0.4074$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2826 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	82.3524	mm
Parallel to Vessel Wall, opening length	d	41.1762	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	21.2500	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N-13-Shell Inle.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9074	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.4375	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.4375	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(7.44, 3)]$
 $= 4.9200$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.91, 4.92)$
 $= 4.9200$ mm

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.5,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	259.5 N./mm ²	Passed
Occasional	:	2.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	29.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A



Govrn. thk, tg = 9.69, tr = 0.41, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 9.69, tr = 0.41, c = 1.5 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2307.51 / 10210.00 = 0.226$

Weld Size Calculations, Description: N-13-Shell Inle

Intermediate Calc. for nozzle/shell Welds Tmin 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.9500 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.3021 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 161.3021 mm

Input Echo, WRC297 Item 1, Description: N-13-Shell Inl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	38.176	mm
Nozzle Thickness	11.0744	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.



Design Internal Pressure	Dp	2307.51	KPa.
Radial Load	P	-163.17	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	30.58	Kg-m.
Longitudinal Moment	Ml	50.97	Kg-m.
Torsional Moment	Mt	61.16	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.447) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(355.75(10.0 - 1.5))}$$

$$= 241.791 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	711.500 mm
Nozzle Outside Diameter	(d) =	60.325 mm
Vessel Thickness used	(T) =	8.500 mm
Nozzle Thickness used	(t) =	9.574 mm
T / t	=	0.888
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.776$$

Nr/P	=	0.139
Mr/P	=	0.229
M0/P	=	0.076
N0/P	=	0.138
MrD/Mc	=	0.292
NrDL/Mc	=	0.132
M0d/Mc	=	0.109
N0DL/Mc	=	-0.049
MrD/Ml	=	0.225
NrDL/Ml	=	0.118
M0D/Ml	=	0.085
N0DL/Ml	=	0.085

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-9	-9	9	9
Outplane Bending (Ml)	-58	58	58	-58
Normal Pressure Stress	94	96	94	96
Outplane Stress Summary	39	138	175	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0

Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-154	154	154	-154
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	-87	160	248	-121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	3	3	-3	-3
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	12	12	12	12
Shear Stress Summary	15	15	9	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	130	168	249	163

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-45	45	45	-45
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	47	47	47	47
Outplane Stress Summary	18	88	101	-8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-120	120	120	-120
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	94	96	94	96
Inplane Stress Summary	-1	180	257	-42

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-3	-3	3	3
Torsional Shear (Mt)	12	12	12	12
Shear Stress Summary	9	9	15	15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Two * Max Shear Stress 27 181 258 48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	-1	0	17	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-29	-29	29	29
Axial Bending (ML)	-89	89	89	-89
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-99	46	139	-73

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	13	13	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	101	53	140	95

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	7	4	7
Hoop Stress Summary	11	14	4	7

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	16	-16	16	-16

Axial Membrane (Mc)	-17	-17	17	17
Axial Bending (Mc)	-73	73	73	-73
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-70	42	111	-68
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	83	44	113	80

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	94	96	94	96	94	96	94	96
Circ. Pl (SUS)	-6	-6	12	12	-6	-6	12	12
Circ. Q (SUS)	-48	48	68	-68	-90	90	150	-150
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	-10	-10	16	16	6	6	0	0
Long. Q (SUS)	-124	124	184	-184	-35	35	55	-55
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	94.0	96.0	94.0	96.0	94.0	96.0	94.0	96.0
Pm+Pl (SUS)	88.2	90.2	106.2	108.2	88.3	90.2	106.2	108.1
Pm+Pl+Q (Total)	130.5	168.4	248.1	162.0	26.9	180.9	257.4	47.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	138.00	Passed
Pm+Pl (SUS)	108.20	207.00	Passed
Pm+Pl+Q (TOTAL)	257.45	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	4	7	4	7	4	7	4	7
Circ. Pl (SUS)	-6	-6	12	12	6	6	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-28	-28	30	30	-16	-16	18	18
Long. Q (SUS)	-73	73	105	-105	-57	57	89	-89
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	4.0	7.0	4.0	7.0	4.0	7.0	4.0	7.0
Pm+Pl (SUS)	26.2	27.3	32.2	32.3	24.3	27.3	20.2	20.3
Pm+Pl+Q (Total)	100.5	51.9	137.5	93.4	82.6	45.0	110.4	79.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	32.30	177.00	Passed
Pm+Pl+Q (TOTAL)	137.53	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N14-Shell Outlet

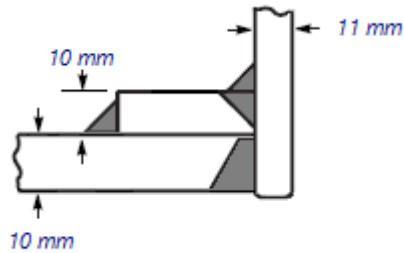
From: 60

Pressure for Reinforcement Calculations	P	2301.269	KPa.
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	1245.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2120.70	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material [Impact Tested]		SA-333 6	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	228.2750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		30.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N14-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	146.329 mm.
Actual Thickness Used in Calculation	10.973 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2301 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 2301)$
 $= 5.9213 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2301 \cdot 74.7 / (118 \cdot 1 - 0.6 \cdot 2301)$
 $= 1.4735 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5132 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.6588 mm
Parallel to Vessel Wall, opening length	d	149.3294 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		21.2500 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

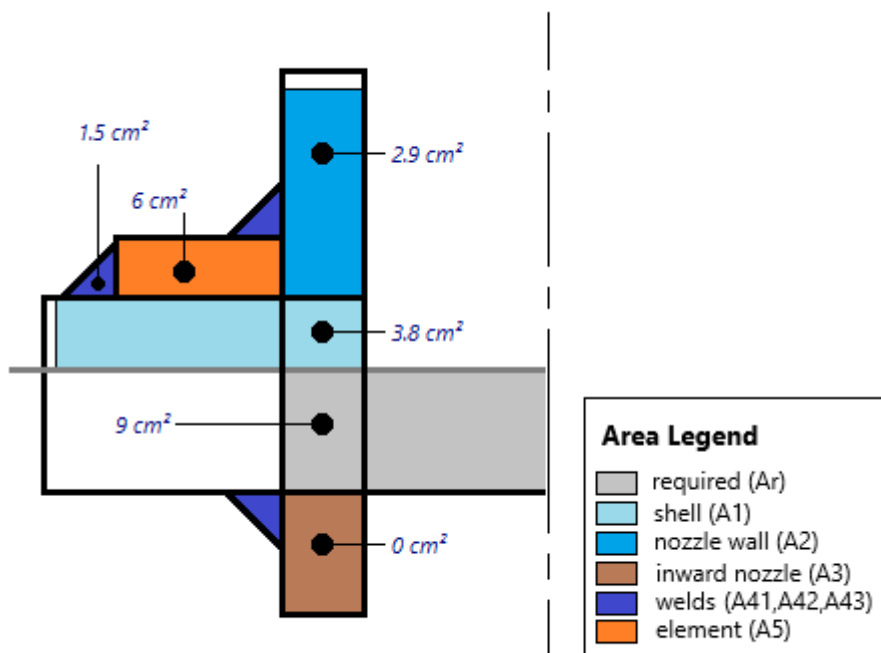


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.005	2.208	NA
Area in Shell	A1	3.780	8.203	NA
Area in Nozzle Wall	A2	2.907	3.256	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.495	1.495	NA
Area in Element	A5	6.000	6.000	NA
TOTAL AREA AVAILABLE	Atot	14.182	18.954	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 176.5019 10.0000 mm
 Based on given Pad Diameter: 228.2750 1.3712 mm
 Based on Shell or Nozzle Thickness: 176.5019 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (149 \cdot 5.92 \cdot 1 + 2 \cdot 9.47 \cdot 5.92 \cdot 1 \cdot (1 - 0.86))$
 $= 9.005 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 149(1 \cdot 8.5 - 1 \cdot 5.92) - 2 \cdot 9.47$
 $(1 \cdot 8.5 - 1 \cdot 5.92) \cdot (1 - 0.86)$
 $= 3.780 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 21.2) \cdot (9.47 - 1.47) \cdot 0.86$



$$= 2.907 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 8^2 \cdot 1$
 $= 1.495 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (228 - 168) \cdot 10 \cdot 1$
 $= 6.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 2.9735 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 3.0000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 7.4213 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 7.4213 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 1.7636 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 7.7200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.72, \max(7.42, 3)]$
 $= 7.4213 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.97, 7.42)$
 $= 7.4213 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.0,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 262.0 N./mm ²	Passed
Occasional	: 8.5,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 17.4,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.6$, $t_r = 1.47$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C
Min Metal Temp. w/o impact per UG-20(f)	-29°C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46°C
Calculated Minimum Design Metal Temperature	-104°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.6$, $t_r = 1.47$, $c = 1.5 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.18$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29°C
--	----------------------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10, tr = 5.92, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.7$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.6, tr = 1.47, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.18$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -46 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $2301.27 / 5110.00 = 0.450$

Weld Size Calculations, Description: N14-Shell Outle

Intermediate Calc. for nozzle/shell Welds Tmin 9.4728 mm
Intermediate Calc. for pad/shell Welds TminPad 8.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	4.2500 = 0.5*TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9 - 3.78 + 2 \cdot 9.47 \cdot 0.86 \cdot (1 \cdot 8.5 - 5.92)) \cdot 138)$
 $= 7939.90 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2.91 + 6 + 1.5 - 0 \cdot 0.86) \cdot 138$
 $= 14637.35 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (2.91 + 0 + 0.86 + (1.38)) \cdot 138$
 $= 7231.46 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (2.91 + 0 + 1.5 + 6 + (1.38)) \cdot 138$
 $= 16574.98 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$

= 15584. Kgf

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 228 \cdot 8 \cdot 0.49 \cdot 138$$

$$= 19779. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 79.4) \cdot (11 - 1.5) \cdot 0.7 \cdot 118$$

$$= 19902. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 10 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (10 - 1.5) \cdot 0.74 \cdot 138$$

$$= 23396. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (19779 + 19902) = 39681 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 27524 + 23396 + 0) = 66504 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (19779 + 23396 + 0) = 43175 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 39681 \text{ Kgf, must exceed } W = 7939 \text{ Kgf} \quad \text{or } W1 = 14637 \text{ Kgf}$$

$$\text{Path 2-2} = 66503 \text{ Kgf, must exceed } W = 7939 \text{ Kgf} \quad \text{or } W2 = 7231 \text{ Kgf}$$

$$\text{Path 3-3} = 43174 \text{ Kgf, must exceed } W = 7939 \text{ Kgf} \quad \text{or } W3 = 16574 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2945 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 10.2635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 170.2635 mm

Input Echo, WRC297 Item 1, Description: N14-Shell Outl:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	700.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	



Nozzle Material	SA-333 6
Nozzle Cold S.I. Allowable	SNmc 118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	228.275 mm
Thickness of Reinforcing Pad	10.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUS tained loads.

Design Internal Pressure	Dp	2301.27 KPa.
Radial Load	P	-407.91 Kgf
Circumferential Shear	Vc	611.87 Kgf
Longitudinal Shear	Vl	611.87 Kgf
Circumferential Moment	Mc	275.24 Kg-m.
Longitudinal Moment	Ml	366.98 Kg-m.
Torsional Moment	Mt	458.73 Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(360.75(10.0 - 1.5))}$$

$$= 351.012 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	721.500 mm
Nozzle Outside Diameter (d)	=	168.275 mm
Vessel Thickness used (T)	=	18.500 mm
Nozzle Thickness used (t)	=	9.473 mm
T / t	=	1.953
d / t	=	17.764

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.457$$

Nr/P	=	0.094
Mr/P	=	0.105
M0/P	=	0.058
N0/P	=	0.149
MrD/Mc	=	0.209
NrDL/Mc	=	0.149
M0d/Mc	=	0.142
N0DL/Mc	=	0.117
MrD/Ml	=	0.150
NrDL/Ml	=	0.106
M0D/Ml	=	0.098
N0DL/Ml	=	0.230

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-36	36	36	-36
Normal Pressure Stress	42	44	42	44



Outplane Stress Summary -2 65 99 19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-56	56	56	-56
Inplane Pressure Stress	21	21	21	21

Inplane Stress Summary -33 64 92 -34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary 6 6 4 4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 34 71 101 55

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	21	21	21	21

Outplane Stress Summary -18 53 72 -15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-58	58	58	-58
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	42	44	42	44

Inplane Stress Summary -14 90 116 -13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	21	90	117	21
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	23	20	23

Hoop Stress Summary	7	10	36	39
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-138	138	138	-138
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-132	113	185	-122
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10

Shear Stress Summary	13	13	8	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	142	115	185	162
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	23	20	23
Hoop Stress Summary	16	19	27	30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-144	144	144	-144
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	-132	124	185	-133

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	8	8	13	13

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	150	125	186	165

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(355.75(10.0 - 1.5))}$$

$$= 349.741 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	711.500 mm
Nozzle Outside Diameter	(d) =	228.275 mm
Vessel Thickness used	(T) =	8.500 mm
Nozzle Thickness used	(t) =	39.473 mm
T / t	=	0.215
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.935$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.059
Mr/P	=	0.108
M0/P	=	0.033
N0/P	=	0.064
MrD/Mc	=	0.268
NrDL/Mc	=	0.092
M0d/Mc	=	0.080
N0DL/Mc	=	0.085
MrD/Ml	=	0.085
NrDL/Ml	=	0.065
M0D/Ml	=	0.024
N0DL/Ml	=	0.079

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	94	96	94	96
Outplane Stress Summary	59	103	157	74

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-111	111	111	-111
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	-39	111	211	-83

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	100	117	212	157

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-78	78	78	-78
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	47	47	47	47
Outplane Stress Summary	-31	104	154	-25

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35

Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-262	262	262	-262
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	94	96	94	96

Inplane Stress Summary	-144	311	411	-184
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	144	311	411	184

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	42	44	42	44	42	44	42	44
Circ. Pl (SUS)	-13	-13	15	15	-5	-5	7	7
Circ. Q (SUS)	-32	32	40	-40	-51	51	65	-65
Long. Pm (SUS)	21	21	21	21	21	21	21	21
Long. Pl (SUS)	-5	-5	7	7	-4	-4	6	6
Long. Q (SUS)	-49	49	63	-63	-35	35	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	42.0	44.0	42.0	44.0	42.0	44.0	42.0	44.0
Pm+Pl (SUS)	29.1	31.1	57.0	59.0	37.0	39.0	49.0	51.0
Pm+Pl+Q (Total)	34.2	70.1	99.0	54.6	20.5	90.4	114.8	21.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	44.00	138.00	Passed
Pm+Pl (SUS)	59.03	207.00	Passed
Pm+Pl+Q (TOTAL)	114.80	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int.	Stress Values at
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Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	94	96	94	96	94	96	94	96
Circ. Pl (SUS)	-14	-14	20	20	-11	-11	17	17
Circ. Q (SUS)	-21	21	41	-41	-227	227	297	-297
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	-11	-11	17	17	-10	-10	16	16
Long. Q (SUS)	-76	76	146	-146	-68	68	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	94.0	96.0	94.0	96.0	94.0	96.0	94.0	96.0
Pm+Pl (SUS)	80.0	82.0	114.0	116.0	83.0	85.0	111.0	113.0
Pm+Pl+Q (Total)	100.0	115.8	210.5	157.3	144.2	312.1	408.2	184.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	138.00	Passed
Pm+Pl (SUS)	116.02	207.00	Passed
Pm+Pl+Q (TOTAL)	408.19	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	23	20	23	20	23	20	23
Circ. Pl (SUS)	-13	-13	15	15	-4	-4	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-123	123	153	-153	-129	129	159	-159
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	20.0	23.0	20.0	23.0	20.0	23.0	20.0	23.0
Pm+Pl (SUS)	17.5	20.4	35.7	38.5	21.4	24.3	27.6	29.8
Pm+Pl+Q (Total)	142.0	114.4	183.4	161.8	150.9	124.6	184.9	164.8

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	118.00	Passed
Pm+Pl (SUS)	38.47	177.00	Passed
Pm+Pl+Q (TOTAL)	184.91	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



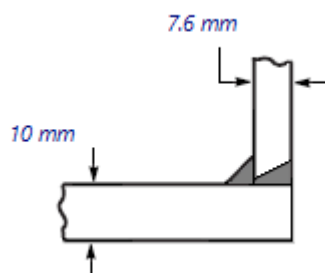
Input, Nozzle Desc: K1A-Liquid Level Transmitter From: 60

Pressure for Reinforcement Calculations	P	2303.328	KPa.
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	1245.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1820.70	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material [Impact Tested]		SA-333 6	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		7.6200	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1A-Liquid Level Transmitter

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 73.660 mm.



Actual Thickness Used in Calculation 7.620 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2303 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 2303)$
= 5.9266 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2303 \cdot 38.3 / (118 \cdot 1 - 0.6 \cdot 2303)$
= 0.7571 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	153.3200	mm
Parallel to Vessel Wall, opening length	d	76.6600	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.3000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A-Liquid Leve.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2571	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.4266	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.4266	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(7.43, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.26, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 7.620 = 6.668$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 6.67, tr = 0.76, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 6.67, tr = 0.76, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2303.33/5110.00 = 0.451

Weld Size Calculations, Description: K1A-Liquid Leve

Intermediate Calc. for nozzle/shell Welds Tmin 6.1200 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2840 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.8341 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 162.8340 mm

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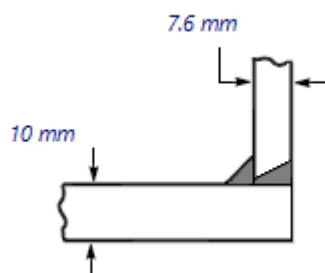
Input, Nozzle Desc: K1B-Liquid Level Transmitter From: 60

Pressure for Reinforcement Calculations	P	2307.513	KPa.
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	700.00	mm
Design Length of Section	L	1245.0000	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1210.70	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material [Impact Tested]		SA-333 6	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		7.6200	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K1B-Liquid Level Transmitter

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 73.660 mm.



Actual Thickness Used in Calculation 7.620 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2308 \cdot 352 / (138 \cdot 1 - 0.6 \cdot 2308)$
= 5.9375 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $2308 \cdot 38.3 / (118 \cdot 1 - 0.6 \cdot 2308)$
= 0.7585 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3552 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	153.3200	mm
Parallel to Vessel Wall, opening length	d	76.6600	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.3000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1B-Liquid Leve.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2585	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.4375	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.4375	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7636	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.3, max(7.44, 3)]
= 6.3000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.26, 6.3)
= 6.3000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 7.620 = 6.668$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 6.67, tr = 0.76, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 6.67, tr = 0.76, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2307.51/5110.00 = 0.452

Weld Size Calculations, Description: K1B-Liquid Leve

Intermediate Calc. for nozzle/shell Welds Tmin 6.1200 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2840 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.8341 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 162.8340 mm

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Input, Nozzle Desc: N9-Tube Outlet

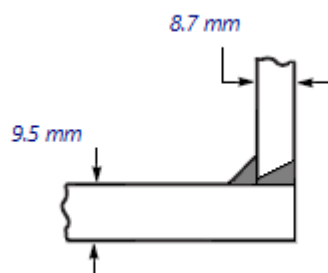
From: 90

Pressure for Reinforcement Calculations	P	2300.003	KPa.
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	80	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	700.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	9.5000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	25.4000	mm
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N9-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 42.850 mm.



Actual Thickness Used in Calculation 8.738 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
= $2300 \cdot 703 \cdot 0.9 / (2 \cdot 138 \cdot 1 \cdot 0.2 \cdot 2300)$
= 5.2603 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
= $2300 \cdot 22.9 / (118 \cdot 1 \cdot 0.6 \cdot 2300)$
= 0.4522 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2826 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	91.6996	mm
Parallel to Vessel Wall, opening length	d	45.8498	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N9-Tube Outlet.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9522	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.3354	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.3354	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7620	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 4.9200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[4.92, max(7.34, 3)]
= 4.9200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.95, 4.92)
= 4.9200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	42.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	252.6 N./mm ²	Passed
Occasional	:	3.1,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	31.9,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.45, c = 1.5 mm, E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.074$, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2300.00/10210.00 = 0.225

Weld Size Calculations, Description: N9-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.0663 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.7139 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 160.2139 mm

Input Echo, WRC107/537 Item 1, Description: N9-Tube Outlet:

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	1260.000	mm
Vessel Thickness	Tv	9.500	mm
Design Temperature	T1	80.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	42.850	mm
Nozzle Thickness	Tn	8.738	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	2300.003	KPa.
Include Pressure Thrust		No	

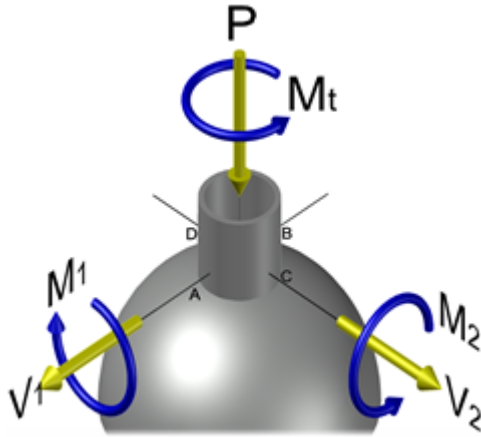
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-163.2	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	30.6	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	51.0	Kg-m.
Torsional Moment (SUS)	Mt	61.2	Kg-m.

WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean} (t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(635.5 (9.5 - 1.5))}$
 $= 295.622 \text{ mm}$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-163.2	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	30.6	Kg-m.
Longitudinal Moment	(ML) M2	51.0	Kg-m.
Torsional Moment	MT	61.2	Kg-m.

Dimensionless Param: U = 0.42 TAU = 5.00 (3.67) RHO = 1.11

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.08233	(A,B,C,D)
$M(x) / P$	SP 2	0.09772	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.16929	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.26948	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.16929	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.26948	(A,B,C,D)
$N(y) * T / P$	SP 2	0.15677	(A,B,C,D)
$M(y) / P$	SP 2	0.04129	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.07914	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.13483	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.07914	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.13483	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of	Stress Intensity Values at
---------	----------------------------



Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Rad.	Bend. P	14.7	-14.7	14.7	-14.7	14.7	-14.7	14.7	-14.7
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-11.1	-11.1	11.1	11.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-106.3	106.3	106.3	-106.3
Rad.	Memb. ML	-18.5	-18.5	18.5	18.5	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-177.1	177.1	177.1	-177.1	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-178.9	146.0	212.4	-171.2	-100.7	82.5	134.1	-107.7
Tang.	Memb. P	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9
Tang.	Bend. P	6.2	-6.2	6.2	-6.2	6.2	-6.2	6.2	-6.2
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-5.2	-5.2	5.2	5.2
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-53.2	53.2	53.2	-53.2
Tang.	Memb. ML	-8.7	-8.7	8.7	8.7	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-88.6	88.6	88.6	-88.6	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-87.2	77.7	107.4	-82.2	-48.3	45.7	68.5	-50.2
	Shear VC	3.3	3.3	-3.3	-3.3	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-3.3	-3.3	3.3	3.3
	Shear MT	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1
Tot. Shear		16.4	16.4	9.8	9.8	9.8	9.8	16.4	16.4
Str. Int.		181.8	149.7	213.3	172.2	102.4	85.0	138.0	112.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	90.8	90.8	90.8	90.8	90.8	90.8	90.8	90.8
Rad.	Pl (SUS)	-16.5	-16.5	20.6	20.6	-9.1	-9.1	13.2	13.2
Rad.	Q (SUS)	-162.4	162.4	191.8	-191.8	-91.6	91.6	120.9	-120.9
Long.	Pm (SUS)	90.8	90.8	90.8	90.8	90.8	90.8	90.8	90.8
Long.	Pl (SUS)	-4.7	-4.7	12.6	12.6	-1.3	-1.3	9.1	9.1
Long.	Q (SUS)	-82.4	82.4	94.8	-94.8	-47.0	47.0	59.4	-59.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	3.3	3.3	-3.3	-3.3	-3.3	-3.3	3.3	3.3
Shear	Q (SUS)	13.1	13.1	13.1	13.1	13.1	13.1	13.1	13.1
Pm (SUS)		90.8	90.8	90.8	90.8	90.8	90.8	90.8	90.8
Pm+Pl (SUS)		86.9	86.9	112.6	112.6	90.7	90.7	105.8	105.8
Pm+Pl+Q (Total)		97.5	240.5	304.1	91.1	56.0	175.8	228.8	66.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	90.79	138.00	Passed
Pm+Pl (SUS)	112.58	207.00	Passed
Pm+Pl+Q (TOTAL)	304.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	2300.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2300.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	10.0000 mm
Shell Internal Corrosion Allowance	cas	1.5000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	700.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	80.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.01 N./mm ²
Shell Allowable Stress at Ambient		138.01 N./mm ²

Channel Description: Bottom Ch Shell-2

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	2300.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	2300.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	10.0000 mm
Channel Internal Corrosion Allowance	cac	1.5000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	700.000 mm
Channel Design Temperature	TEMPC	150.00 °C
Channel Flange Matl		SA-516 60

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02100
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	316
Tube Wall Thickness	et	2.0000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	1400.00 mm
Straight Tube Length (bet. inner tubsht faces) L		1250.00 mm
Design Temperature of the Tubes		150.00 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	214.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	35.0000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.3000 mm
Groove Weld Leg	ag	1.3000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	327.826 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	3034.1 cm ²
Perimeter of Tube Layout	Cp	2025.0 mm



Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 150.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 75.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm
 Tubesheet Outside Diameter A 890.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No
 Area of the Untubed Lanes AL 0.0 cm²

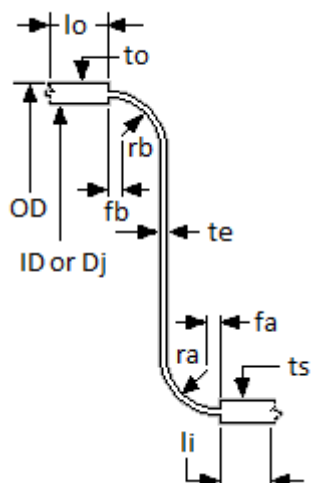
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 750.000 mm
 Tube End condition corresponding to Span (l) k 1.00
 Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Channel Side Min. Design Pressure Ptd,min -103.4190 KPa.
 Mean Shell Metal Temp. along Shell len. Tsm -38.00 °C
 Mean Tube Metal Temp. along Tube length Ttm -38.00 °C

Expansion Joint Type Thick Flanged/Flued
 Thick Expansion Joint Analysis Method TEMA 8th/9th/10th Ed.
 Expansion Joint Inside Diameter ID or Dj 900.000 mm
 Outside Diameter of Expansion Joint OD 920.000 mm
 Expansion Joint Corrosion Allowance Caj 1.5000 mm
 Exp Jt Material SA-350 LF1

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Exp Jt Material UNS Number K03009
 Exp Jt Allowable Stress at Temperature Soe 118.00 N./mm²
 Exp Jt Allowable Stress At Ambient Sae 118.00 N./mm²
 Wall thickness of Expansion Joint te 10.0000 mm
 Length of Shell Cylinder li 1250.0000 mm
 Length of Straight Flange, Joint ID fa 30.0000 mm
 Radius of Knuckle, Joint ID ra 30.0000 mm
 Length of Straight Flange, Joint OD fb 30.0000 mm
 Radius of Knuckle, Joint OD rb 30.0000 mm
 Desired Cycle life 750
 Print Detailed Expansion Joint Calculations No
 Num. of Flexible Shell Elemts. (1 Convol = 2 Fse) 4



Junction Stress Reduction option

None

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Channel
Flange Outside Diameter	A 890.000 mm
Flange Inside Diameter	B 700.000 mm
Flange Face Outside Diameter	Fod 775.000 mm
Flange Face Inside Diameter	Fid 700.000 mm
Gasket Outside Diameter	Go 764.000 mm
Gasket Inside Diameter	Gi 732.000 mm
Small end Hub thk.	g0 10.0000 mm
Large end Hub thk.	g1 15.0000 mm
Gasket Factor,	m 3.00
Gasket Design Seating Stress	y 68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a
Column for Gasket Seating	Code Column II
Gasket Thickness	tg 3.1750 mm
Full face Gasket Flange Option	Program Selects

Bolting Information:

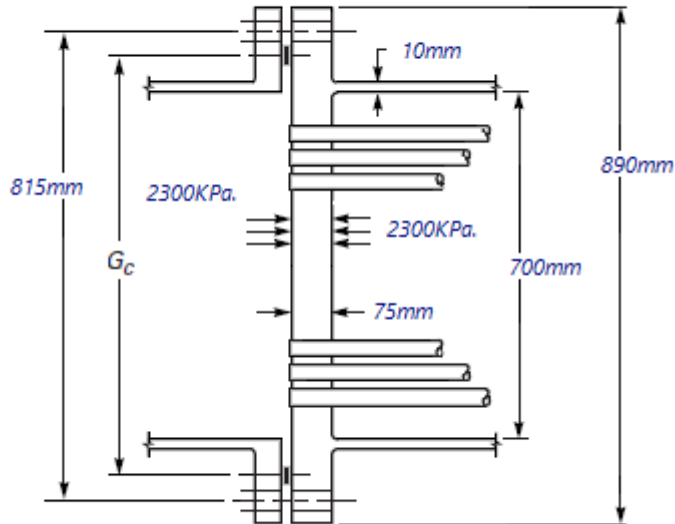
Diameter of Bolt Circle	C 815.000 mm
Nominal Bolt Diameter	dB 27.0000 mm
Thread Series	TEMA Metric Thread
Number of Bolts	n 28
Bolt Material	SA-320 L7
Bolt Allowable Stress At Temperature	Sb 172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa 172.00 N./mm ²
Weld between Flange and Shell/Channel	0.0000 mm

Tubesheet Integral with	Shell
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 43.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2023

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	-38.0 °C	0.0000106686 /°C
Elastic Mod. at Design Temperature	80.0 °C	0.19933E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	-38.0 °C	0.20658E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	-38.0 °C	0.0000106686 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Metal Temp. along Len	-38.0 °C	0.20658E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Expan. Jt. - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Metal Temperature	-38.0 °C	0.20658E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (2300 \cdot (25/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 2300)$
 $= 0.2418 + 0.0000 = 0.2418 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 150.00 °C
Elastic Modulus for Material 199807168.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.0000	25.00	1250.00	12.50	50.0000	0.0070400	122.42

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122) / (3 \cdot 12.5) = 13058 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6352	25.00	1250.00	39.36	50.0000	0.0007101	70.95

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 71) / (3 \cdot 39.4) = 2404 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6352 mm
Allowable Internal Pressure at Corroded thickness	20169.40 KPa.
Required Internal Design Pressure	2300.00 KPa.
Allowable External Pressure at Corroded thickness	13057.60 KPa.
Required External Design Pressure	2403.42 KPa.
Required Thickness due to Shell Side pressure	0.6352 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2023

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	16.000 mm
Basic Gasket Width,	$b_0 = N / 2$	8.000 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	7.125 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	749.750 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Compute the tube expansion depth ratio $[\rho]$:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/75$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Compute the effective tube hole diameter $[d^*]$:

$= \max(d_t - 2 \cdot t_t (E_t / E) \cdot (S_t / S) \cdot (\rho); d_t - 2 \cdot t_t)$
 $= \max(25 - 2 \cdot 2 (195062688 / 195062688) \cdot (118 / 138) \cdot (0); 25 - 2 \cdot 2)$
 $= 25.0000 \text{ mm}$

Compute the equivalent outer tube limit circle diameter $[D_o]$:

$= 2 \cdot r_o + d_t$
 $= 2 \cdot 328 + 25$
 $= 680.652 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:



$$\begin{aligned} &= (p - dt) / p \\ &= (35 - 25) / 35 \\ &= 0.286 \end{aligned}$$

Compute the equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 681 / 2 \\ &= 340.3260 \text{ mm} \end{aligned}$$

Compute the effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ &= 35 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100; 4 \cdot 681 \cdot 35) / (\pi \cdot 681^2))} \\ &= 35.0000 \text{ mm} \end{aligned}$$

Compute the effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (35 - 25) / 35 \\ &= 0.2857 \end{aligned}$$

Compute the ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 350 / 340 \\ &= 1.0284 \end{aligned}$$

Compute the ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 375 / 340 \\ &= 1.1015 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt \cdot (dt - 2 \cdot tt) / (2 \cdot ao)^2 \\ &= 1 - 316 \cdot (25 - 2 \cdot 2) / (2 \cdot 340)^2 \\ &= 0.6992 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt \cdot (dt / (2 \cdot ao))^2 \\ &= 1 - 316 \cdot (25 / (2 \cdot 340))^2 \\ &= 0.5737 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max(hg - CATC; 0) \text{ (For pressure only cases)} \\ &= \max(8 - 0; 0) \\ &= 8.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Determine the Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts \cdot (Ds + ts) \cdot Es / L \\ &= \pi \cdot 10 \cdot (700 + 10) \cdot 199328640 / 1250 \\ &= 362714.1250 \text{ Kg/mm} \end{aligned}$$

Determine the Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot tt \cdot (Dt - tt) \cdot Et / L \\ &= \pi \cdot 2 \cdot (25 - 2) \cdot 195062688 / 1250 \\ &= 2299.6853 \text{ Kg/mm} \end{aligned}$$

Compute the Stiffness Factor [Ks,t]:

$$\begin{aligned} &= Ks / (Nt \cdot Kt) \\ &= 362714 / (316 \cdot 2300) \\ &= 0.49912 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned} &= 1 / (1 + Ks / Kj) \\ &= 1 / (1 + 362714 / 10576) \\ &= 0.0283 \text{ (= 1 if No Exp. Jt.)} \end{aligned}$$

Compute shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts) \cdot ts) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((700 + 10) \cdot 10) \\ &= 0.0216 \text{ 1/mm} \end{aligned}$$



Determine shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.022 \cdot 199328640 \cdot 10^3 / (6(1 - 0.3^2))$$

$$= 787595968.0000 \text{ KPa} \cdot \text{mm}^2$$

Determine shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(700 + 10) \cdot 787595968 / 75^3 \cdot (1 + 75 \cdot 0.022 + (75 \cdot 0.022)^2 / 2)$$

$$= 31231724.0000 \text{ KPa}$$

Determine shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 700^2 / (4 \cdot 199328640 \cdot 10) \cdot (1 - 700 \cdot 0.3 / (2(700 + 10)))$$

$$= 0.52365\text{E-}01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

SME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 2.142857 ; \mu^* = 0.285714 ; E^* = 51202248 \text{ KPa}$$

$$E^*/E = 0.262491 ; \nu^* = 0.364266$$

Note: As h/p (2.143) is > 2, data values for h/p = 2 were used.

Compute the Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.36^2)316 \cdot 195062688 \cdot 2$$

$$(25 - 2)340^2) / (51202248 \cdot 1250 \cdot 75^3))^{1/4}$$

$$= 3.9889$$

Values from Table 4.18.3

$$Z_d = 0.024132 ; Z_v = 0.063429 ; Z_m = 0.368771$$

$$Z_a = 0.675501\text{E}+01 ; Z_w = 0.063429$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Compute the diameter ratio [K]:

$$= A / D_o$$

$$= 890/681$$

$$= 1.3076$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.36) / 51202248 \cdot (31231724 + 0 +$$

$$195062688 \cdot \ln(1.31))$$

$$= 1.0373$$

Compute parameter [φ]:

$$= (1 + \nu^*)F$$

$$= (1 + 0.36)1.04$$

$$= 1.4151$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.03 - 1 - 1.42 \cdot 0.063) / (1 + 1.42 \cdot 0.37)$$

$$= -0.040302$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot Xa^4$$

$$= (0.024 + -0.04 \cdot 0.063) / 2 \cdot 3.99^4$$

$$= 2.7312$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot Xa^4$$

$$= (0.063 + -0.04 \cdot 0.37) / 2 \cdot 3.99^4$$

$$= 6.1479$$



Compute parameter [U]:

$$= (Z_w + (\rho_s - 1)Z_m)X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.063 + (1.03 - 1)0.37)3.99^4 / (1 + 1.42 \cdot 0.37)$$

$$= 12.2957$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (750 - 815) / 681 = -0.096$$

Compute Parameter [y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.03 \cdot 787595968 \cdot 0.022 \cdot 0.052364651 (1 + 75 \cdot 0.022)$$

$$= 2395.7502 \text{ mm}^2$$

Calculate parameter [w*s]:

$$= a_o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - w_s$$

$$= 340^2 (1.03^2 - 1) \cdot (1.03 - 1) / 4 - 2396$$

$$= -2348.2920 \text{ mm}^2$$

Calculate parameter [wc]:

$$= 0 \text{ because } \beta_C \text{ is equal to zero}$$

Calculate parameter [w*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - w_c$$

$$= 340^2 [(1.1^2 + 1) \cdot (1.1 - 1) / 4 - (1.03 - 1) / 2] - 0$$

$$= 4859.9609 \text{ mm}^2$$

Compute the pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Compute the pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Compute the pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2 / K_{st} \cdot (D_s / D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 2300 \{ 0.57 + 2(1 - 0.57)0.3 + [2 / 0.5 \cdot (700 / 681)^2]0.3 - [(1.03^2 - 1) / (0.028 \cdot 0.5)] - [(1 - 0.028) / (2 \cdot 0.028 \cdot 0.5)] [(900^2 - (700)^2) / 681^2] \}$$

$$= -59123.0742 \text{ KPa.}$$

Compute the pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1 / (J \cdot K_{st})] P_t$$

$$= [0.7 + 2(1 - 0.7)0.3 + 1 / (0.028 \cdot 0.5)] 2300$$

$$= 164662.6875 \text{ KPa.}$$

Compute the pressure [Pγ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 316 \cdot 2300 \cdot 0 / (3.14 \cdot 340^2)$$

$$= 0.0000 \text{ KPa.}$$

Compute the pressure [Pw]:

$$= -\gamma_b \cdot U \cdot W_* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.096 \cdot 12.3 \cdot 127166 / (2 \cdot \pi \cdot 340^2)$$

$$= 2019.8105 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U / a_o^2) \cdot (w_* s \cdot P_s - w_* c \cdot P_t)$$

$$= -(12.3 / 340^2) \cdot (-2348 \cdot 2300 - 4860 \cdot 2300)$$

$$= 1760.0342 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U / a_o^2) \cdot (w_s \cdot P_* s - w_c \cdot P_* c)$$



$$= (12.3/340^2) \cdot (2396 \cdot 0 - 0 \cdot 0) \\ = 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot \\ (P's - P't + Py + Pw + Pw + Prim) \\ = 0.2833E-01 \cdot 0.5 / (1 + 0.028 \cdot 0.5(2.73 + (1.03 - \\ 1)6.15)) \cdot (-59123 - 164663 + 0 + 0 + 2020 + 1760) \\ = -2988.4460 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega s \cdot P \cdot s - \omega c \cdot P \cdot c)) + \\ W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ = [((-2348 \cdot 2300 - 4860 \cdot 2300) - \\ (2396 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\ 127166 \cdot -0.096 / (2 \cdot \pi)] / (1 + 1.42 \cdot 0.37) \\ = -2385.8044 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ = -0.04 + 2 \cdot -2386 / (-2988 \cdot 340^2) \\ = 0.09488$$

Fm Value from Table 4.18.3 = 0.096066

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.096 / 0.29) \cdot (2 \cdot 340 / (75 - 8))^2 \cdot -2988 \\ = -155.5614 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.093 / 0.29) \cdot (2 \cdot 340 / (64.8 - 8))^2 \cdot -2959 \\ = -206.9861 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc \\ = 64.8 + 0 + 0 \\ = 64.7604 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ = (2 \cdot 0.29 \cdot 75 / 340) \cdot 110 \\ = 13.9024 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taujimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp) \\ = (3034 \cdot -2501) / (0.29 \cdot 12.5 \cdot 2025) \\ = -104.9327 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + Cats + Catc \\ = 12.5 + 0 + 0 \\ = 12.5022 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(HreqB; HreqS) \\ = \max(64.8; 12.5) \\ = 64.7604 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:



The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -1.07, Ftmax = 3.82$$

First Extreme Tube Axial Stress from among all the tubes [$\sigma t1$]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\ &= ((2300 \cdot 0.57 - 2300 \cdot 0.7) - (-2988) \cdot -1.07) / \\ &\quad (0.7 - 0.57)) \\ &= -27.8558 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [$\sigma t2$]:

$$\begin{aligned} &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\ &= ((2300 \cdot 0.57 - 2300 \cdot 0.7) - (-2988) \cdot 3.82) / \\ &\quad (0.7 - 0.57)) \\ &= 88.5850 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma t,max$]:

$$= \max(| \sigma t,1 | ; | \sigma t,2 |) = 88.6 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma t,A$]:

$$= Sot = 117.9979 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned} &= \max((3.25 - 0.25(Z_d + Q3 \cdot Z_w) Xa^4); 1.25) \\ &= \max((3.25 - 0.25(0.024 + 0.095 \cdot 0.063) 3.99^4); 1.25) \\ &= 1.3417 \text{ (Should be } \leq 2 \text{)} \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ &= (\sqrt{ 25^2 + (25 - 2 \cdot 2)^2 }) / 4 \\ &= 8.1624 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 195062688 / 214) } \\ &= 134.1405 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1 \cdot 750 / 8.16 \\ &= 91.8846 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (Sy, t / Fs) \cdot (1 - Ft / (2 \cdot Ct)) \\ &= (214 / 1.34) \cdot (1 - 91.9 / (2 \cdot 134)) \\ &= 104.871 \text{ N./mm}^2 \text{ (Never greater than Sot)} \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma t,max \cdot \text{Tube Area} \\ &= 88.6 \cdot 1.45 \\ &= 1305.39 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (do - t) t \\ &= \pi (25 - 2)^2 \\ &= 1.445 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (do - t) Sa \\ &= \pi \cdot 2 (25 - 2) 118 \\ &= 1738.814 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$



$$= 0.85 \cdot \pi \cdot 1.3 (25 + 0.67 \cdot 1.3) 118$$
$$= 1080.6146 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t)$$
$$= 118/118$$
$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$
$$= 0.55 \cdot \pi \cdot 1.3 (25 + 0.67 \cdot 1.3) 118$$
$$= 699.2213 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\frac{\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d}}{0.75 \cdot d_o} \right) - 0.75 \cdot d_o$$
$$= 2 \left(\frac{\sqrt{(0.75 \cdot 25)^2 + 1.07 \cdot 2 \cdot (25 - 2) 1 \cdot 1}}{0.75 \cdot 25} \right) - 0.75 \cdot 25$$
$$= 2.539 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2696 mm

Minimum Required Groove Weld Leg [agr]: 1.2696 mm

Tube-Tubesheet Jt allowable, 1739 is $\geq$ tube strength 1739 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / (D_s + t_s) t_s \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / (D_s + t_s) t_s$$
$$= 340^2 / ((700 + 10) 10) \cdot [-2988 + (1.03^2 - 1)(2300 - 2300)] + 350^2 \cdot 2300 / ((700 + 10) 10)$$
$$= -9.0678 \text{ N./mm}^2$$

Note: The Shell is under Axial Compression, checking the Compressive Stress

For the Shell Attached to the Tubesheet, Factor A per UG-23:

$$= 0.125 / (S R_o / t_s) = 0.125 / (360 / 10) = 0.0035$$

Longitudinal Compressive Allowable Stress [S_s, b]:

$$= 114.3649 \text{ N./mm}^2$$

Note: The Shell Axial Compressive stress is within limits.

Displacement used in Expansion Joint Analysis [ΔM_s]:

$$= \sigma_s, m (T_s (D_s + T_s) \pi) / (j \cdot K_s) - \pi \cdot D_s^2 \cdot P_s \cdot v_s / (2 \cdot K_s) + \pi / 8 (D_j^2 - D_s^2) P_s / K_j$$
$$= -9.07 (10 (700 + 10) \pi) / (0.028 \cdot 362714) + \pi \cdot 700^2 \cdot 2300 \cdot 0.3 / (2 \cdot 362714) + \pi / 8 (900^2 - 700^2) 2300 / 10576$$
$$= -2.00689292 - 0.14931259 + 2.786726$$
$$= 0.630520 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
For a given Shell thickness of 10 mm, the minimum Shell length adjacent to the tubesheet should be 150.599 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / (D_s + t_s) t_s [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / (D_s + t_s) t_s$$
$$= 340^2 / ((700 + 10) 10) [-2988 + (1.03^2 - 1)(2300 - 2300)] + 350^2 \cdot 2300 / ((700 + 10) 10)$$



= -9.0678 N./mm²

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 787595968 / 10^2 \{ 0.022 [0.052364651 \cdot 2300 + 350^2 \cdot 0 / (1993286272)] + 6(1 - 0.36^2) / (51202248) (340 / 75)^3 (1 + 75 \cdot 0.022 / 2) [-2988 (0.063 + 0.37 \cdot -0.04) + 2 / 340^2 \cdot 0.37 \cdot -2386] \}$$

$$= -115.9754 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore | \sigma_{s,m} | + | \sigma_{s,b} | \leq 1.5 \cdot S_s$$

$$\therefore | -9.07 | + | -116 | \leq 207 \text{ N./mm}^2$$

$$\therefore 125, \text{ must be } \leq 207 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	155.6	<= 207.0 N./mm ²	Ok
Tubesheet Shear Stress	20.9	<= 110.4 N./mm ²	Ok
Maximum Tube Stress	88.6	<= 118.0 N./mm ²	Ok
Minimum Tube Stress (Buckling)	-27.9	<= -104.9 N./mm ²	Ok
Maximum Force on any one Tube	1305.4	<= 1738.8 Kg _f	Ok
Axial Membrane Stress in Shell	9.1	<= 138.0 N./mm ²	Ok
Axial Compressive Shell Stress	-9.1	<= -114.4 N./mm ²	Ok
Shell Stress (jt. inter.)	125.0	<= 207.0 N./mm ²	Ok

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	64.760	75.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.270	1.300	Ok
Tube-Tubesheet Groove Weld Leg :	1.270	1.300	Ok

Note: Computing Effective Tubesheet Pressures acting on the Thick Exp. Jt.
 (The Elastic Modulus is at the Mean Metal Temperature.)

TEMA RCB-7.161/A.151 J Factor for Thermal Expansion

$$J = S_j \cdot L / (S_j \cdot L + P_i (d_o - t_s) \cdot t_s \cdot E_s) = 0.0274$$

TEMA RCB-7.161/A.151 K Factor for Thermal Exp. 0.4884

TEMA RCB-7.161/A.151 F_q Factor for Thermal Exp. 3.7

TEMA RCB-7.161/A.151 Differential Exp. / Length 0.00E+00

TEMA RCB-7.161/A.151 Equivalent Differential Thermal Exp. Pressure [PD]:

$$= 4 \cdot J \cdot E_s \cdot T_s \cdot D_{LL} / (D_{OS} - 3 \cdot T_s) (1 + J \cdot R_K \cdot F_Q)$$

$$= 4 \cdot 0.027 \cdot 206597 \cdot 10 \cdot 0 / (720 - 3 \cdot 10) (1 + 0.027 \cdot 0.49 \cdot 3.73)$$

$$= 0.000 \text{ KPa.}$$

TEMA RCB 7.163/A.153 Effective Shellside Design Pressure:

TEMA RCB-7.163/A.153 FF_s Factor for Shellside Design 0.5969

$$PSP1 = 0.4 \cdot J (1.5 + K \cdot (1.5 + FF_s))$$

$$= 0.4 \cdot 0.027 (1.5 + 0.49 \cdot (1.5 + 0.6))$$

$$PSP2 = ((0.5 - J / 2) \cdot (D_J^2 / G^2 - 1))$$

$$= ((0.5 - 0.027 / 2) \cdot (900^2 / 700^2 - 1))$$

$$PSP3 = 1 + J \cdot K \cdot F_Q$$

$$= 1 + 0.027 \cdot 0.49 \cdot 3.73$$

$$\begin{aligned} \text{PSP} &= \text{PS} ((\text{PSP1} - \text{PSP2}) / \text{PSP3}) \\ &= 2300 ((0.028 - 0.32) / 1.05) \end{aligned}$$

$$\text{PSP} = -635.269 \text{ KPa.}$$

TEMA RCB 7.164/A.154 Effective Tubeside Design Pressure:
 TEMA RCB-7.164/A.154, FFt Factor for Tubeside Design 0.7156

$$\begin{aligned} \text{PTP1} &= 1.0 + 0.4 \cdot \text{J} \cdot \text{K} \cdot (1.5 + \text{FFT}) \\ &= 1.0 + 0.4 \cdot 0.027 \cdot 0.49 \cdot (1.5 + 0.72) \end{aligned}$$

$$\begin{aligned} \text{PTP2} &= 1.0 + \text{J} \cdot \text{K} \cdot \text{FQ} \\ &= 1.0 + 0.027 \cdot 0.49 \cdot 3.73 \end{aligned}$$

$$\begin{aligned} \text{PTP} &= \text{PT} \cdot \text{PTP1} / \text{PTP2} \\ &= 2300 \cdot 1.01 / 1.05 \end{aligned}$$

$$\text{PTP} = 2216.802 \text{ KPa.}$$

Expansion Joint Stress Summary (per TEMA and ASME Sec VIII Div 1 App-5):

Analysis of Shellside Pressure (Ps),(N./mm²):

	Inside Junction		Outside Junction	
	Actual	Allowed	Actual	Allowed
Annular Element	80	176	118	176
Annular Ring at R =		400.480	-72	176
Cyl. at point Y	92	176	...	176
Cyl. at point L	72	176	...	176
Max. Cycle Stress	97	(Fatigue)	...	(Fatigue)
Max. Cycle Life	999999	Cycles	Unlimited	Cycles

Analysis of Tubeside Pressure (Pt),(N./mm²):

	Inside Junction		Outside Junction	
	Actual	Allowed	Actual	Allowed
Annular Element	-41	176	35	176
Annular Ring at R =		355.000	41	176
Cyl. at point Y	1	176	...	176
Cyl. at point L	4	176	...	176
Max. Cycle Stress	43	(Fatigue)	...	(Fatigue)
Max. Cycle Life	999999	Cycles	Unlimited	Cycles

Analysis of Shellside + Tubeside Pressure (Ps + Pt),(N./mm²):

	Inside Junction		Outside Junction	
	Actual	Allowed	Actual	Allowed
Annular Element	38	176	153	176
Annular Ring at R =		394.752	-77	176
Cyl. at point Y	93	176	...	176
Cyl. at point L	77	176	...	176
Max. Cycle Stress	53	(Fatigue)	...	(Fatigue)
Max. Cycle Life	999999	Cycles	Unlimited	Cycles

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2023

Results for 16 Load Cases:

--Reqd. Thk. + CA			---- Tubesheet Stresses			Case	Pass/	
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail



D1uc	58.418	31.155	124	207	15	110	Ps+Pt-Th	D1	Ok
D2uc	21.600	...	28	207	5	110	Ps+Pt-Th	D2	Ok
D3uc	64.760	...	156	207	21	110	Ps+Pt-Th	D3	Ok
D4uc	25.000	...	4	207	1	110	Ps+Pt-Th	D4	Ok
O1uc	32.468	...	111	438	15	110	Ps+Pt+Th	O1	Ok
O2uc	25.368	...	77	438	5	110	Ps+Pt+Th	O2	Ok
O3uc	36.044	...	135	438	21	110	Ps+Pt+Th	O3	Ok
O4uc	23.819	...	61	438	...	110	Ps+Pt+Th	O4	Ok
D1c	61.553	31.155	135	207	16	110	Ps+Pt-Th-c	D1	Ok
D2c	24.586	...	31	207	5	110	Ps+Pt-Th-c	D2	Ok
D3c	67.905	...	169	207	22	110	Ps+Pt-Th-c	D3	Ok
D4c	13.168	...	4	207	1	110	Ps+Pt-Th-c	D4	Ok
O1c	38.401	...	125	438	16	110	Ps+Pt+Th-c	O1	Ok
O2c	31.457	...	87	438	6	110	Ps+Pt+Th-c	O2	Ok
O3c	41.838	...	152	438	22	110	Ps+Pt+Th-c	O3	Ok
O4c	29.288	...	69	438	...	110	Ps+Pt+Th-c	O4	Ok

Max: 67.9053 31.155 mm 0.815 0.203 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Pass	Fail
D1uc	2	138	...	...	:	...	...	...	...	:	Ok	
D2uc	11	138	-11	-114	:	...	...	...	...	:	Ok	
D3uc	9	138	-9	-114	:	...	...	...	...	:	Ok	
D4uc	...	138	...	-114	:	...	...	...	...	:	Ok	
O1uc	2	486	...	...	:	...	...	...	...	:	Ok	
O2uc	10	486	-10	-114	:	...	...	...	...	:	Ok	
O3uc	9	486	-9	-114	:	...	...	...	...	:	Ok	
O4uc	1	486	...	...	:	...	...	...	...	:	Ok	
D1c	1	138	...	...	:	...	...	...	...	:	Ok	
D2c	13	138	-13	-111	:	...	...	...	...	:	Ok	
D3c	12	138	-12	-111	:	...	...	...	...	:	Ok	
D4c	...	138	...	-111	:	...	...	...	...	:	Ok	
O1c	2	486	...	...	:	...	...	...	...	:	Ok	
O2c	13	486	-13	-111	:	...	...	...	...	:	Ok	
O3c	12	486	-12	-111	:	...	...	...	...	:	Ok	
O4c	1	486	...	...	:	...	...	...	...	:	Ok	
Max RATIO	0.098			0.121	:	...			...	:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass



Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	57	117	-33	-111	838	1738	191	207	...	...	Ok
D2uc	30	117	...	...	439	1738	90	207	...	...	Ok
D3uc	89	117	-28	-104	1305	1738	125	207	...	...	Ok
D4uc	2	117	-2	-85	29	1738	6	207	...	...	Ok
O1uc	61	235	-35	-112	896	3477	210	486	...	...	Ok
O2uc	49	235	-3	-112	721	3477	20	486	...	...	Ok
O3uc	93	235	-30	-112	1364	3477	144	486	...	...	Ok
O4uc	17	235	-8	-70	253	3477	83	486	...	...	Ok
D1c	62	117	-34	-112	914	1738	202	207	...	...	Ok
D2c	32	117	...	...	469	1738	112	207	...	...	Ok
D3c	96	117	-29	-112	1414	1738	120	207	...	...	Ok
D4c	2	117	-2	-89	31	1738	6	207	...	...	Ok
O1c	66	235	-36	-112	979	3477	222	486	...	...	Ok
O2c	53	235	-3	-112	783	3477	17	486	...	...	Ok
O3c	100	235	-31	-112	1479	3477	140	486	...	...	Ok
O4c	19	235	-9	-70	283	3477	89	486	...	...	Ok
Max RATIO	0.813		0.317		0.813		0.976		...		

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	67.905	75.000	Ok
Tubesheet thickness flanged extension :	31.155	43.000	Ok
Tube thickness :	0.635	2.000	Ok
Tube-tubesheet fillet weld leg :	1.270	1.300	Ok
Tube-tubesheet groove weld leg :	1.270	1.300	Ok

Min Shell length of thk, (10.000) adj. to tubesheet: 150.599 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Displacements between Inner Tubesheet Faces:

Maximum Linear Expansion, Pressure + Thermal : 0.7298 mm
 Maximum Linear Expansion, Pressure only : 0.6957 mm
 Minimum Linear Expansion, Pressure only : -0.0143 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	3527.71	1.000	13798.86	1.000
Tubesheet Shear Stress	15631.21	1.000	40790.91	1.000
Tube Tensile Stress	4376.78	1.000	7998.50	1.000
Tube Compressive Stress	7592.53	1.000	No Calc	No Calc
Tube-Tubesheet Joint load	4376.78	1.000	7998.50	1.000
Shell Stress (Axial, Junction)	2357.53	1.000	3444.45	0.766
Tube Pressure Stress	20169.39	1.000	13057.59	1.000
Expansion Joint Stress	9714.73	1.000	2874.13	1.000
Tubesheet Extension Stress	3675.79	...	No Calc	No Calc
Minimum MAWP	2357.53		2874.13	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	3817.76	1.000	14839.45	1.000
Tubesheet Shear Stress	16529.55	1.000	45464.58	1.000
Tube Tensile Stress	4758.68	1.000	8522.07	1.000
Tube Compressive Stress	8226.75	1.000	No Calc	No Calc
Tube-Tubesheet Joint load	4758.69	1.000	8522.07	1.000
Shell Stress (Axial, Junction)	2541.21	1.000	5617.16	1.000
Tube Pressure Stress	20169.39	1.000	13084.31	1.000
Expansion Joint Stress	9714.73	1.000	3444.45	1.000



Tubesheet Extension Stress	3675.79	...	No Calc	No Calc
Minimum MAPnc	2541.21		3444.45	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.3858	0.4329	...	...	0.3858	0.4329
2	0.2318	0.2485	...	...	0.2318	0.2485
3	0.6305	0.6957	...	...	0.6305	0.6957
4	-0.0129	-0.0143	...	...	-0.0129	-0.0143
5	0.0099	0.0099	0.4155	0.4670	0.4155	0.4670
6	0.1829	0.1847	0.3734	0.4107	0.3734	0.4107
7	0.1928	0.1946	0.6601	0.7298	0.6601	0.7298
8	...	...	0.1287	0.1479	0.1287	0.1479

Summary of Expansion Jt Results TEMA/ASME App-5, (worst case),N./mm²:

Category	Max. Stress	Allowable	Location (junction)	Ld Case Tsht	ExpJt	Pass Fail
Annul. Elm.	175.	177.	Outside	D3c	Ps+Pt	Pass
Cyl. at Y	109.	177.	Inside	D3c	Ps+Pt	Pass
Cyl. at L	90.	177.	Inside	D3c	Ps+Pt	Pass

Min. Cycles v/s Desired 999999 750 Inside 03c Ps+Pt Pass

UnCorroded Expansion Joint Axial Spring rate: 10576 Kg/mm
 Corroded Expansion Joint Axial Spring rate : 6809 Kg/mm

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, shell thickness))
 = max(75/4, min(75, 10))
 = 18.750 mm

Thickness Ratio = 0.91, Temperature Reduction per Fig. UCS 66.1 = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
 = tubesheet thickness/4
 = 75/4
 = 18.750 mm

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(2300/2541, 2300/3444)
 = 0.905